

Genetic Algorithm for Gas Turbine Health Indicator Estimation

Experiment Framework on a single measurment.

This notebook implements a **Genetic Algorithm (GA)** to solve the **inverse problem** in gas turbine engine health monitoring.

Systematic Experiments with Multiple Scenarios

This notebook includes:

1. **All original plots** (convergence, 2D/3D search space, ablation)
 2. **Multi-scenario experiments** (1-4 contexts × 1-3 sensors)
 3. **Clean and noisy data** comparison
 4. **Uses existing original and noisy CSV file**
-

For details about the problem and the parameter tuning please refer to :

- the `readme file`
- The `technical documentation` inside the Reports folder

Note : Update the `ga_config.py` if you have new configuration for the GA

```
In [7]: import sys
        from pathlib import Path

        # Add project root to sys.path
        project_root = Path().resolve().parent
        sys.path.append(str(project_root))
```

```
In [8]: project_root
```

```
Out[8]: PosixPath('/home/ensai/Documents/dataviz-course/SDSPROJECT/TO_PUSH/TO_PUSH_TRUE/scenarios/V3SAFRAN_ENSAI_2025/TAKEOFFgenetic_algorithm/simulationv1')
```

```
In [9]: # include GA pachages
        from functions.GA_single_experiment import *

        from functions.ga_config import *
```

```
In [ ]:
```

```
In [10]: %matplotlib inline
         import matplotlib.pyplot as plt
         from mpl_toolkits.mplot3d import Axes3D
         import numpy as np
         import pandas as pd
         from typing import List, Dict, Tuple, Optional
         from dataclasses import dataclass
         import time
         import os
         import pickle
         import warnings
         from datetime import datetime
         import copy
```

```
warnings.filterwarnings('ignore')

# Simulator imports
from odsmr.sensors import HPC_Tout, HP_Nmech, HPC_Tin, LPT_Tin, Fuel_flow, HPC_Pout_s
from odsmr.predefined_flight_conditions import Cruise_DeckSMR, Takeoff_DeckSMR, Climb
from odsmr.generation_functions import decksmr_lforall
from odsmr.constants import ROOT_OPENDECK, STATE_LABELS, STATE_BOUNDS

from scipy.stats import qmc

np.random.seed(42)
print("Imports OK")
```

Imports OK

```
In [11]: print("="*80)
print("ALL 10 HEALTH INDICATORS")
print("="*80)
print(f"{'Index':<6} {'Label':<35} {'Lower':>10} {'Upper':>10}")
print("-"*65)
for i, label in enumerate(STATE_LABELS):
    bounds = STATE_BOUNDS[label]
    print(f"{'i':<6} {'label':<35} {'bounds[0]:>10.3f} {'bounds[1]:>10.3f}")
```

=====

ALL 10 HEALTH INDICATORS

=====

Index	Label	Lower	Upper
0	deg_CmpBst_s_mapEff_in	-0.050	0.000
1	deg_CmpBst_s_mapWc_in	-0.050	0.030
2	deg_CmpFan_s_mapEff_in	-0.050	0.000
3	deg_CmpFan_s_mapWc_in	-0.050	0.030
4	deg_CmpH_s_mapEff_in	-0.050	0.000
5	deg_CmpH_s_mapWc_in	-0.050	0.030
6	deg_TrbH_s_mapEff_in	-0.050	0.000
7	deg_TrbH_s_mapWc_in	-0.050	0.050
8	deg_TrbL_s_mapEff_in	-0.050	0.000
9	deg_TrbL_s_mapWc_in	-0.050	0.050

1. Configuration of the simulator

Note: Update the `ga_config.py` if you have new configuration for the GA

```
In [12]: print(f"Contexts available: {ALL_CONTEXTS}")
print(f"Sensors available: {ALL_SENSORS}")
print(f"Indicators to estimate: {[INDICATOR_SHORT_NAMES[i] for i in INDICATORS_TO_EST]})
```

Contexts available: ['TAKEOFF']
Sensors available: ['HPC_Tin', 'LPT_Tin', 'HPC_Pout_st']
Indicators to estimate: ['Fan Wc', 'HPC Eff', 'HPT Eff']

```
In [13]: print (SENSOR_SCENARIOS)

[['HPC_Tin'], ['HPC_Tin', 'LPT_Tin'], ['HPC_Tin', 'LPT_Tin', 'HPC_Pout_st']]
```

```
In [14]: print(f"Population: {config.population_size}, Elitism: {config.elitism_count}, Tournamen
```

Population: 150, Elitism: 7, Tournament: 15

Load the data

```
In [15]: # Load clean data
df_clean = pd.read_csv(CLEAN_DATA_PATH)
print(f"Loaded {len(df_clean)} rows from {CLEAN_DATA_PATH}")
```

Loaded 1000 rows from ../data/synthetic_data.csv

```
In [ ]:
```

```
In [16]: # LOAD noisy version of data
df_noisy = pd.read_csv (NOISY_DATA_PATH)
print(f"Loaded {len(df_clean)} rows from {NOISY_DATA_PATH}")

# Show noise effect
print("\nNoise effect sample (row 0, CRUISE):")
for s in ALL_SENSORS:
    col = f'CRUISE_DECKSMR{s}'
    clean_val = df_clean.loc[0, col]
    noisy_val = df_noisy.loc[0, col]
    diff = noisy_val - clean_val
    print(f"  {s}: clean={clean_val:.4f}, noisy={noisy_val:.4f}, noise={diff:.4f}")
```

Loaded 1000 rows from ../data/synthetic_data_noisy.csv

Noise effect sample (row 0, CRUISE):

HPC_Tin: clean=336.0642, noisy=336.0353, noise=-0.0289

LPT_Tin: clean=1123.0515, noisy=1123.1817, noise=0.1302

HPC_Pout_st: clean=1377933.0900, noisy=1377498.8378, noise=-434.2522

```
In [ ]:
```

2. GA Configuration and Class

Note: Update the `ga_config.py` if you want new configuration for the GA

3. Run All Experiments

Single measurement estimation + Ablation study

```
In [17]: %%time
# Run all experiments
output_dir, summary_df = run_all_experiments(
    df_clean=df_clean,
    df_noisy=df_noisy,
    n_test_rows=N_TEST_ROWS,
    row_to_use =ROW_TO_USE ,

    output_base_dir=OUTPUT_BASE_DIR
)
```

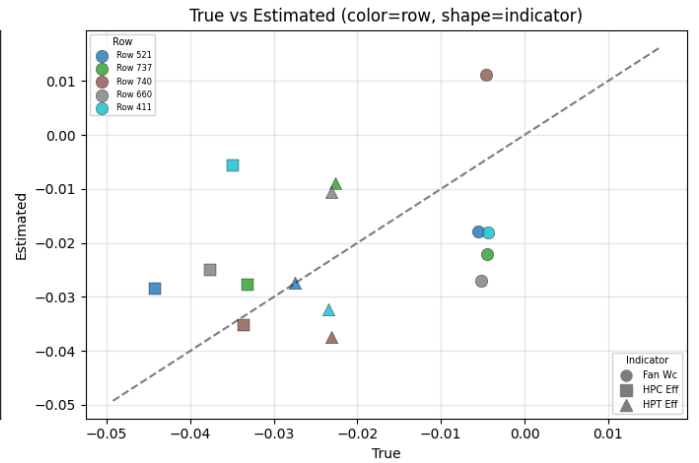
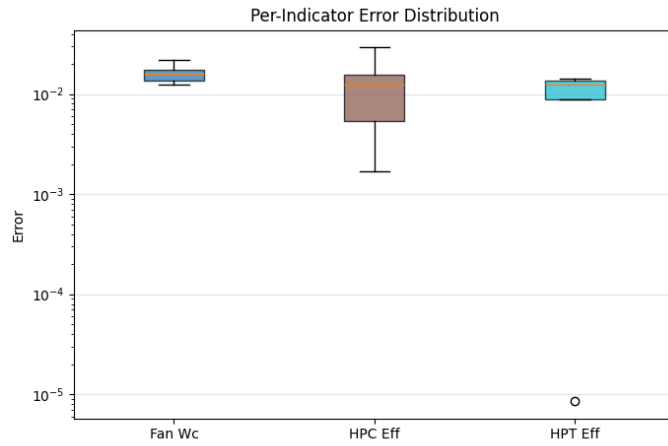
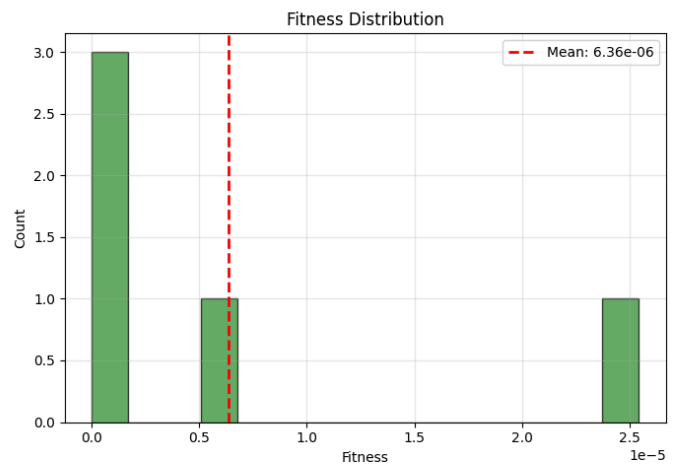
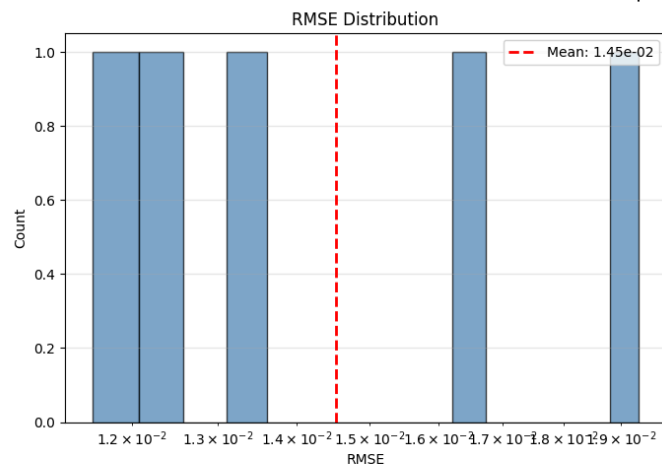
=====

Running experiments on CLEAN data

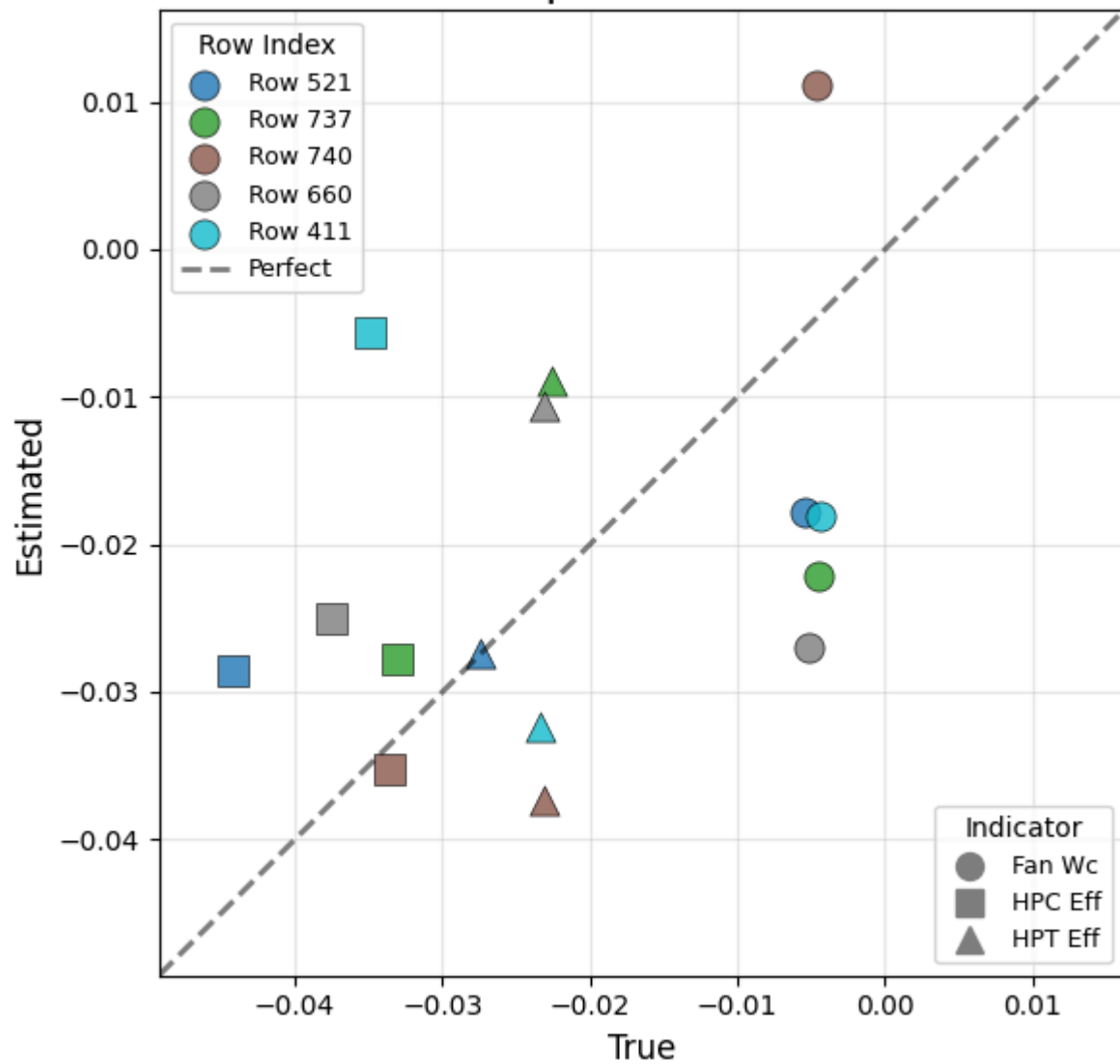
=====

lctx_1sensors: ['TAKEOFF'] × ['HPC_Tin']
1 measurements → 3 indicators

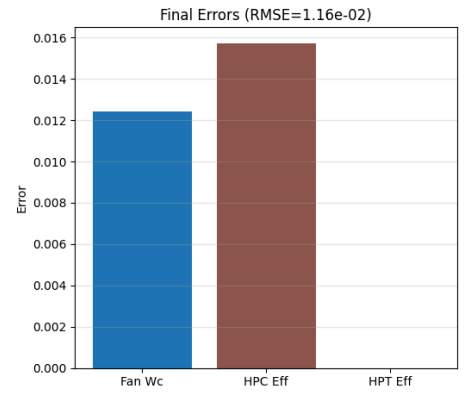
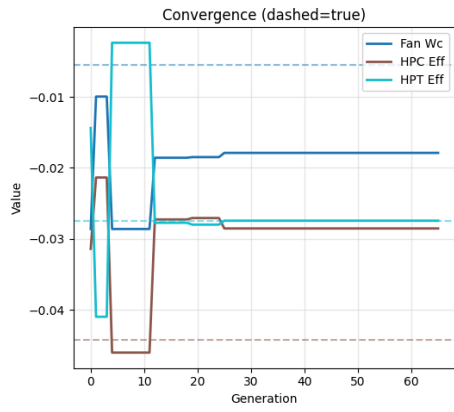
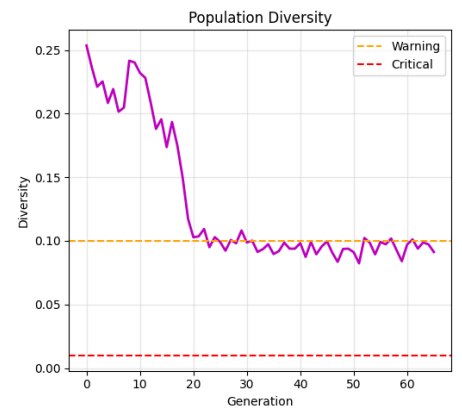
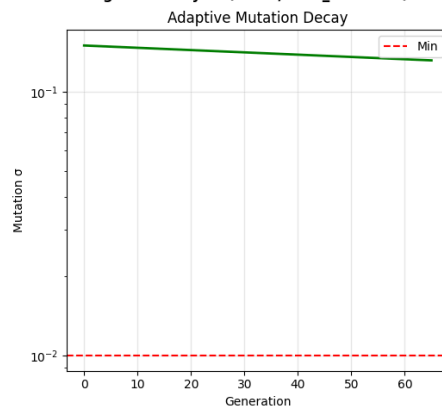
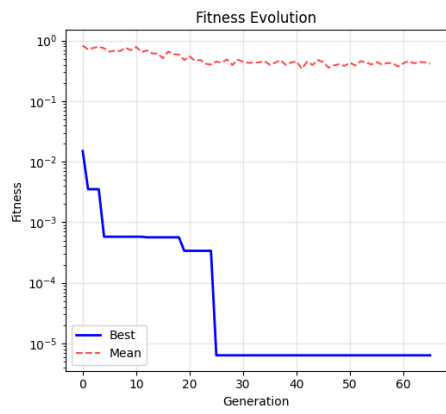
Clean | 1 ctx × 1 sens



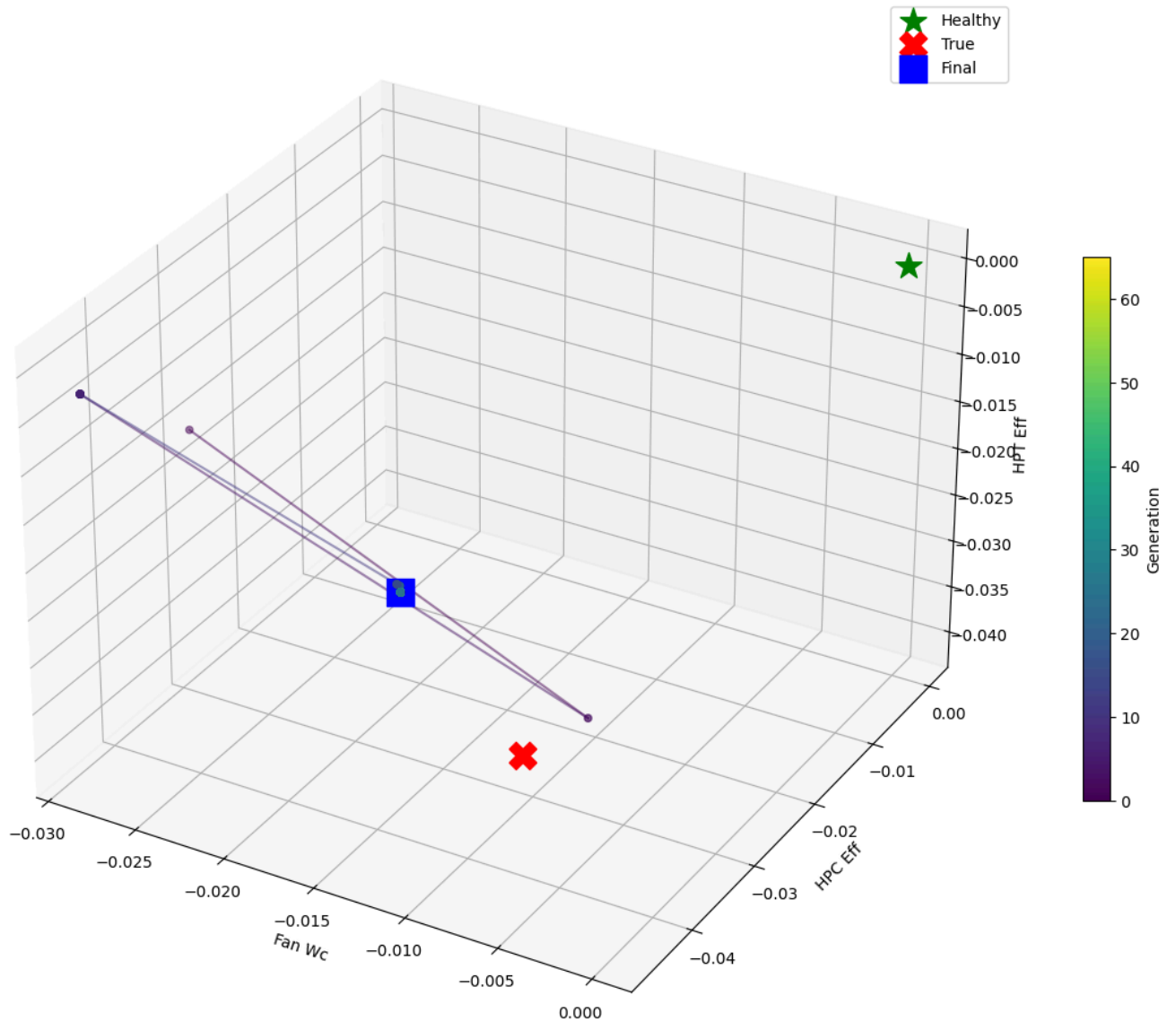
Clean | 1 ctx × 1 sens



Convergence Analysis (clean, 1ctx_1sensors)



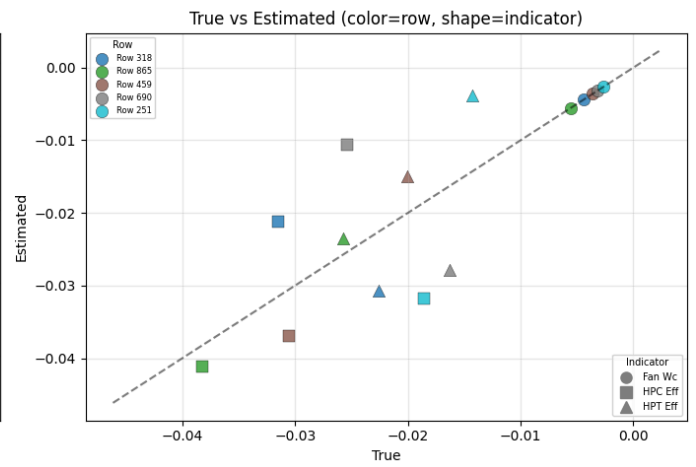
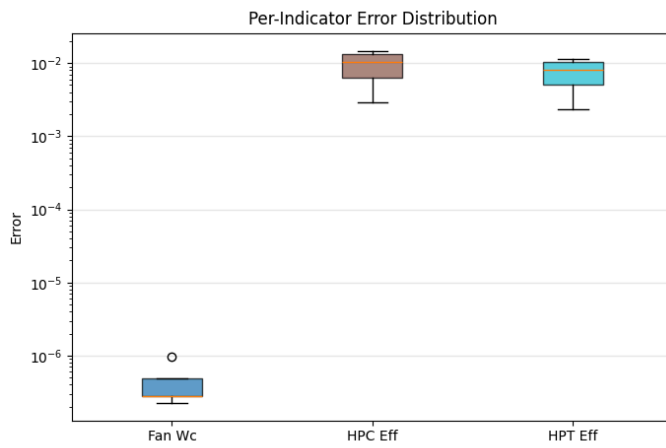
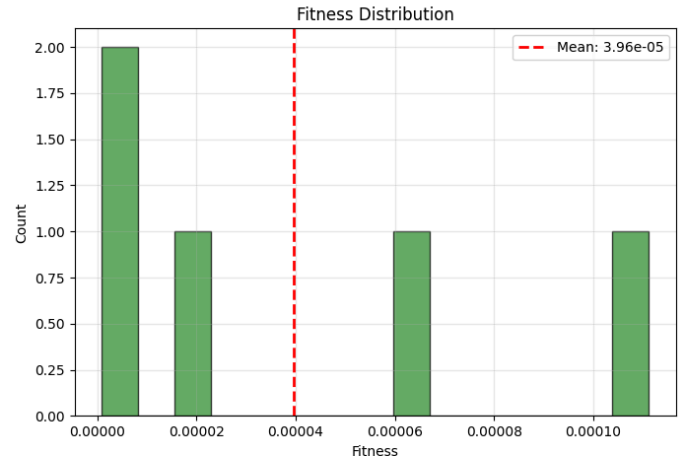
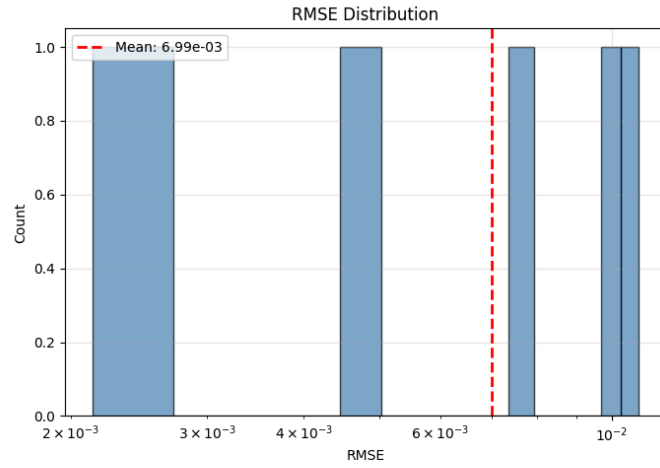
3D Search Trajectory (clean, 1ctx_1sensors)



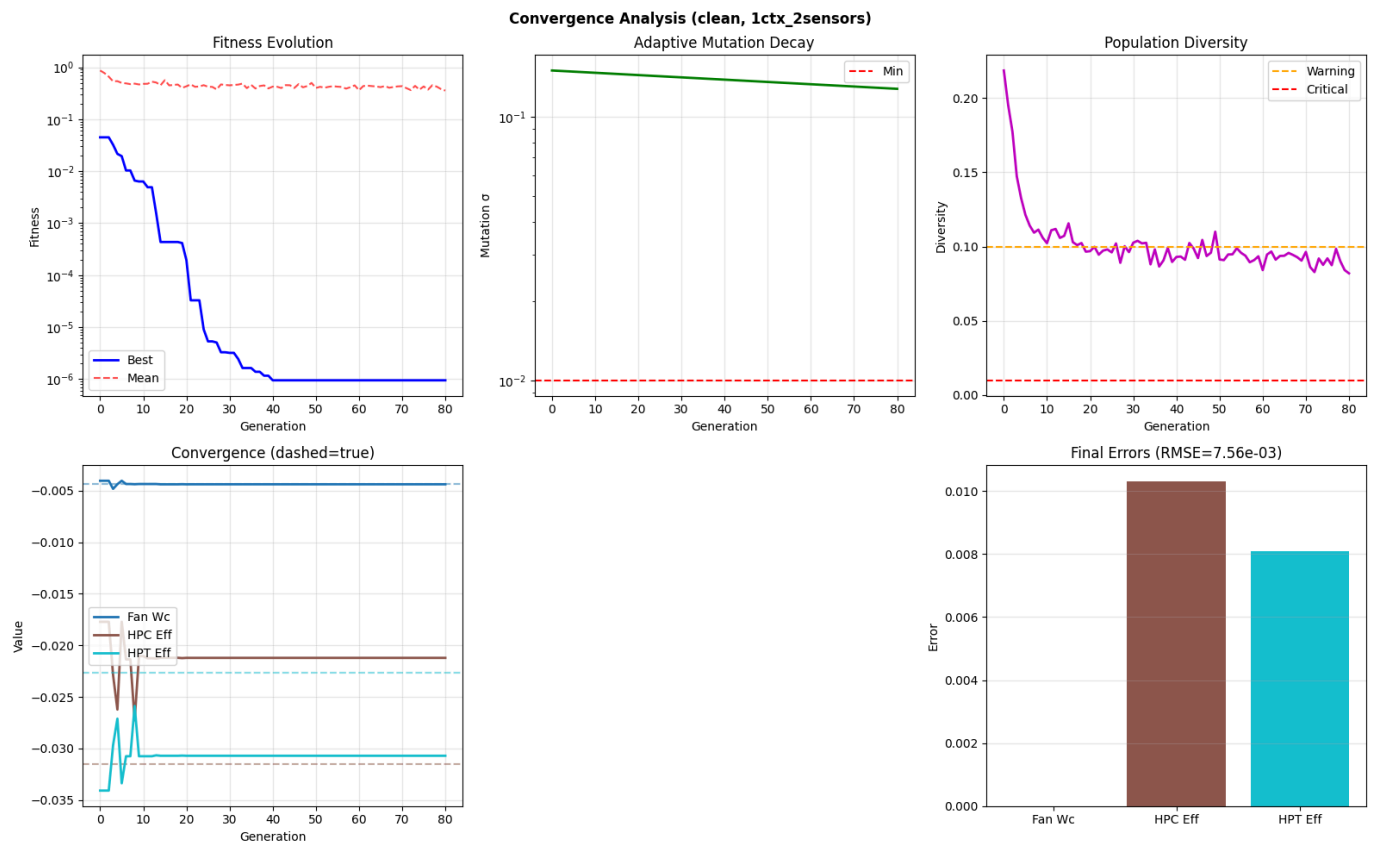
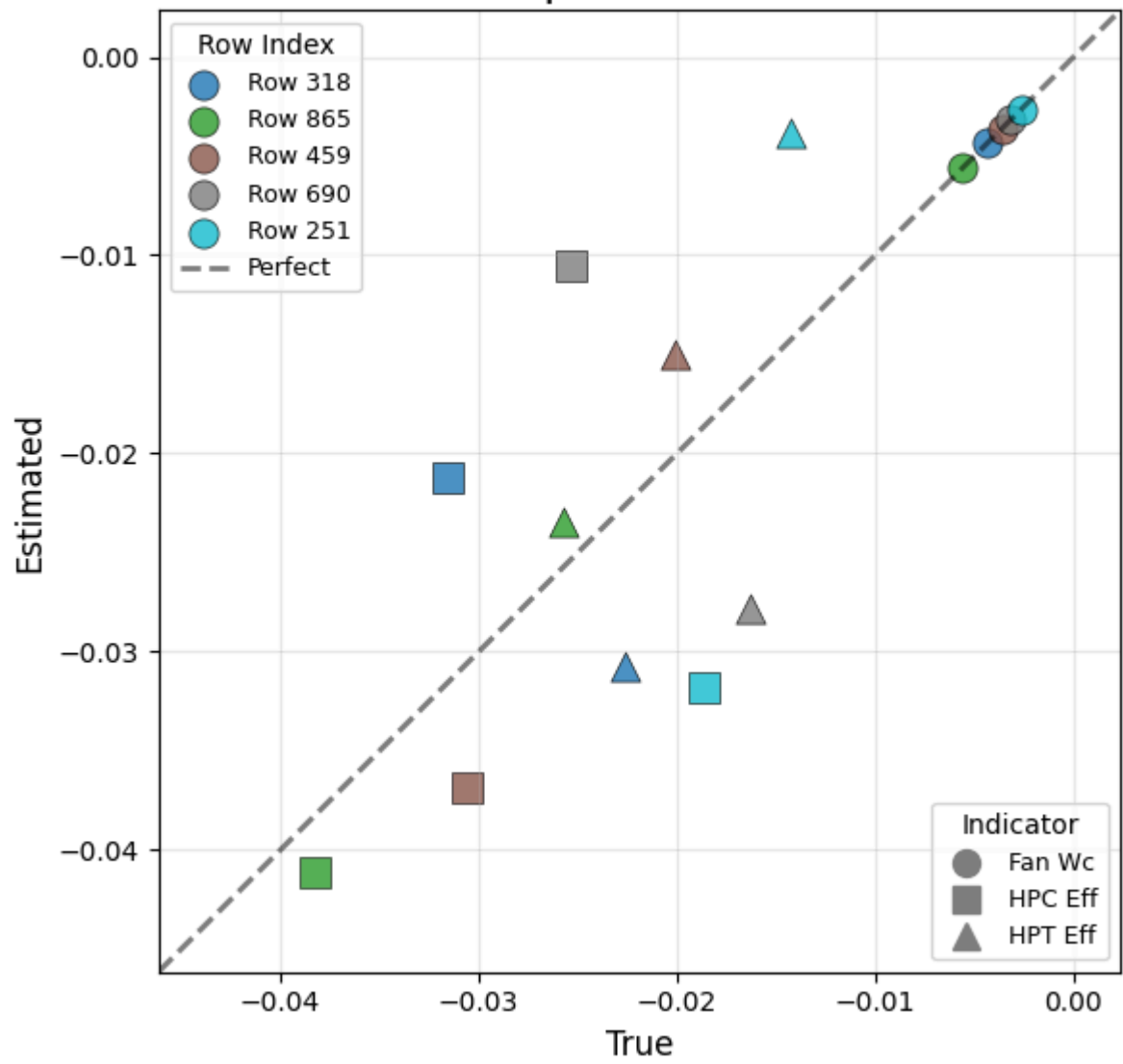
Mean RMSE: $1.45\text{e-}02 \pm 3.21\text{e-}03$
Mean Time: $1839.86\text{s} \pm 1237.30\text{s}$

lctx_2sensors: ['TAKEOFF'] × ['HPC_Tin', 'LPT_Tin']
2 measurements → 3 indicators

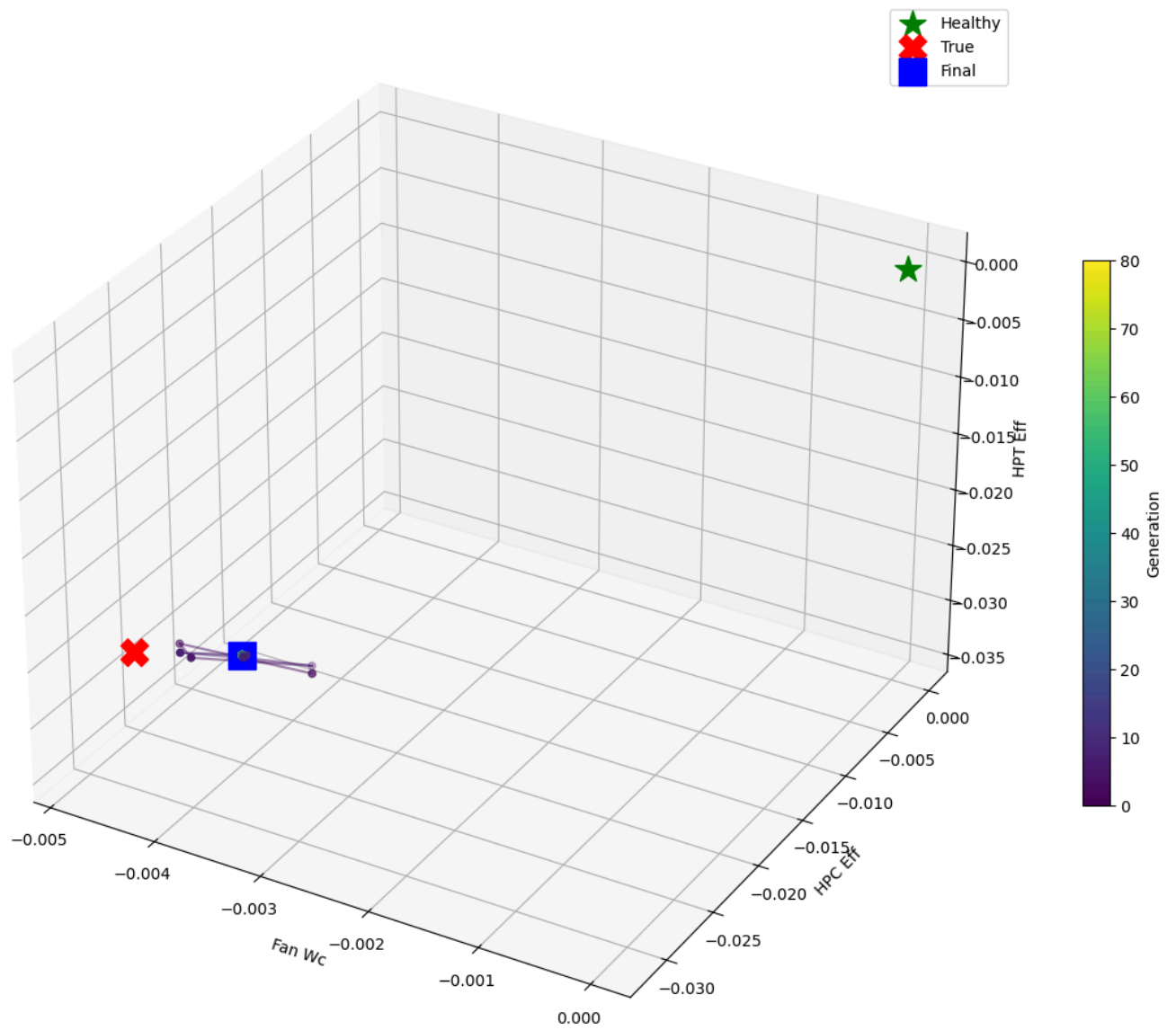
Clean | 1 ctx × 2 sens



Clean | 1 ctx × 2 sens



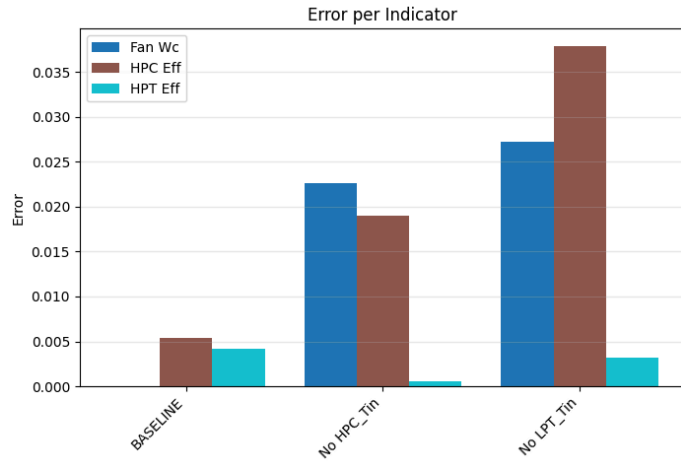
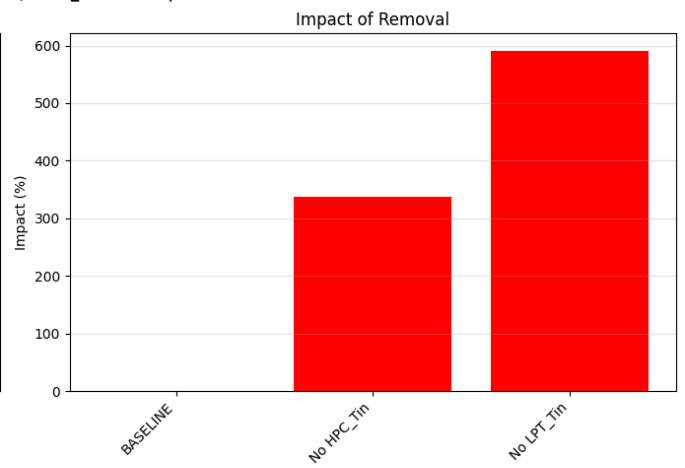
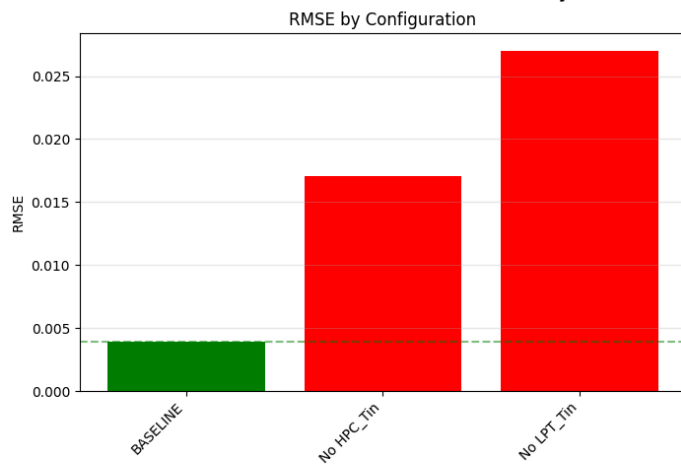
3D Search Trajectory (clean, 1ctx_2sensors)



Running ablation study...
Running BASELINE...
RMSE: 3.91e-03

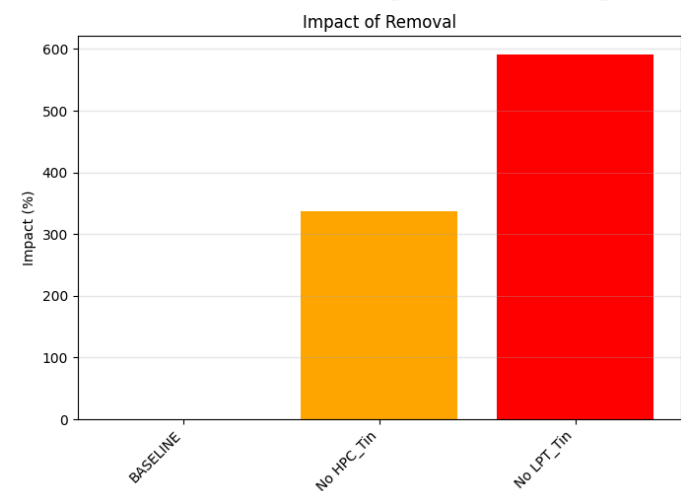
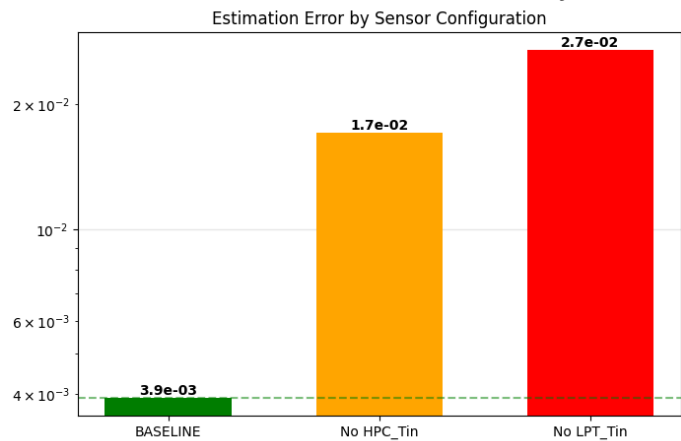
Sensor ablation:
Without HPC_Tin... RMSE: 1.71e-02 (+336.5%)
Without LPT_Tin... RMSE: 2.70e-02 (+591.4%)

Ablation Study Results (clean, 1ctx_2sensors | row 53)



INTERPRETATION:
 BASELINE: RMSE = 3.91e-03
 HPC_Tin: HIGH IMPORTANCE (+336.5%)
 LPT_Tin: HIGH IMPORTANCE (+591.4%)
 Legend:
 DETRIMENTAL: Removing HELPS
 NEUTRAL: Little effect
 HIGH IMPORTANCE: Removing HURTS

Ablation Study Results (clean, 1ctx_2sensors | row 53)



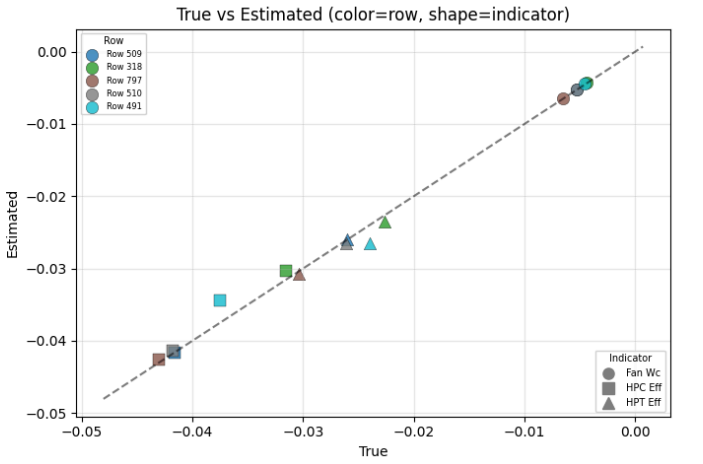
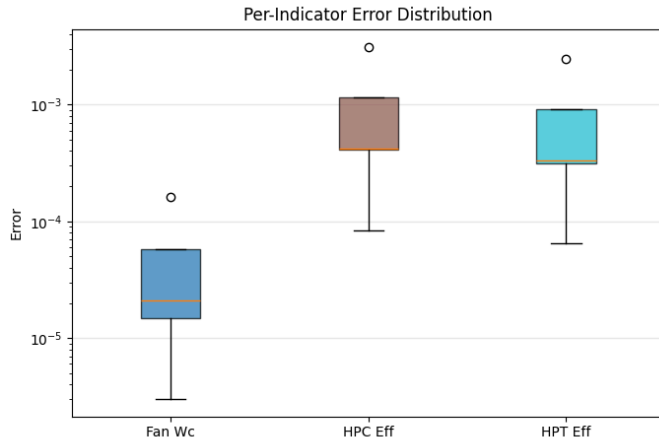
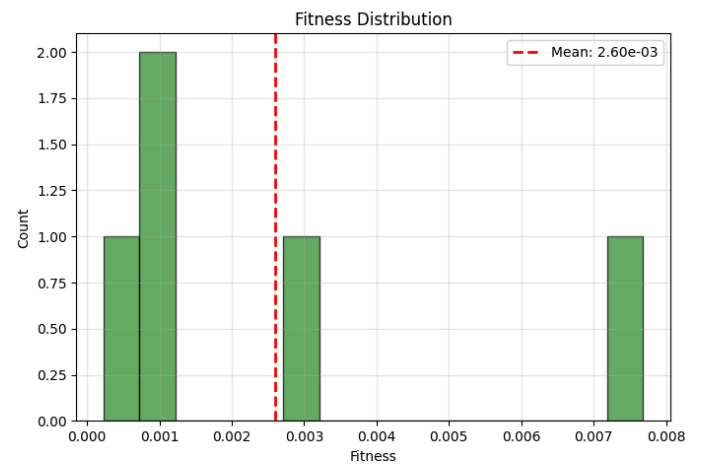
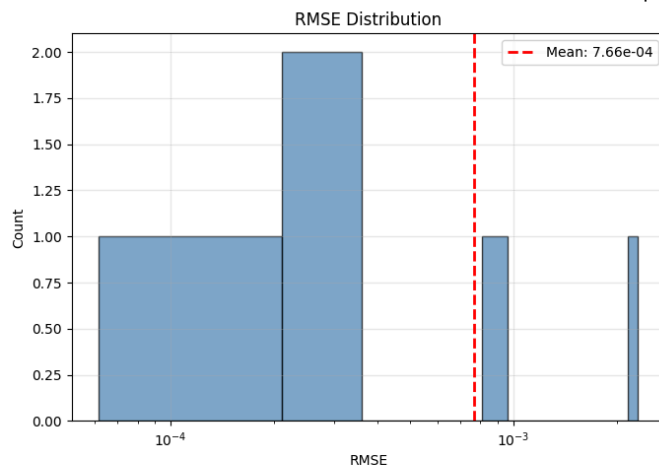
INTERPRETATION:
 BASELINE: RMSE = 3.91e-03
 HPC_Tin: HIGH IMPORTANCE (+336.5%)
 LPT_Tin: HIGH IMPORTANCE (+591.4%)
 Legend:
 DETRIMENTAL: Removing HELPS
 NEUTRAL: Little effect
 HIGH IMPORTANCE: Removing HURTS

Mean RMSE: 6.99e-03 ± 3.59e-03

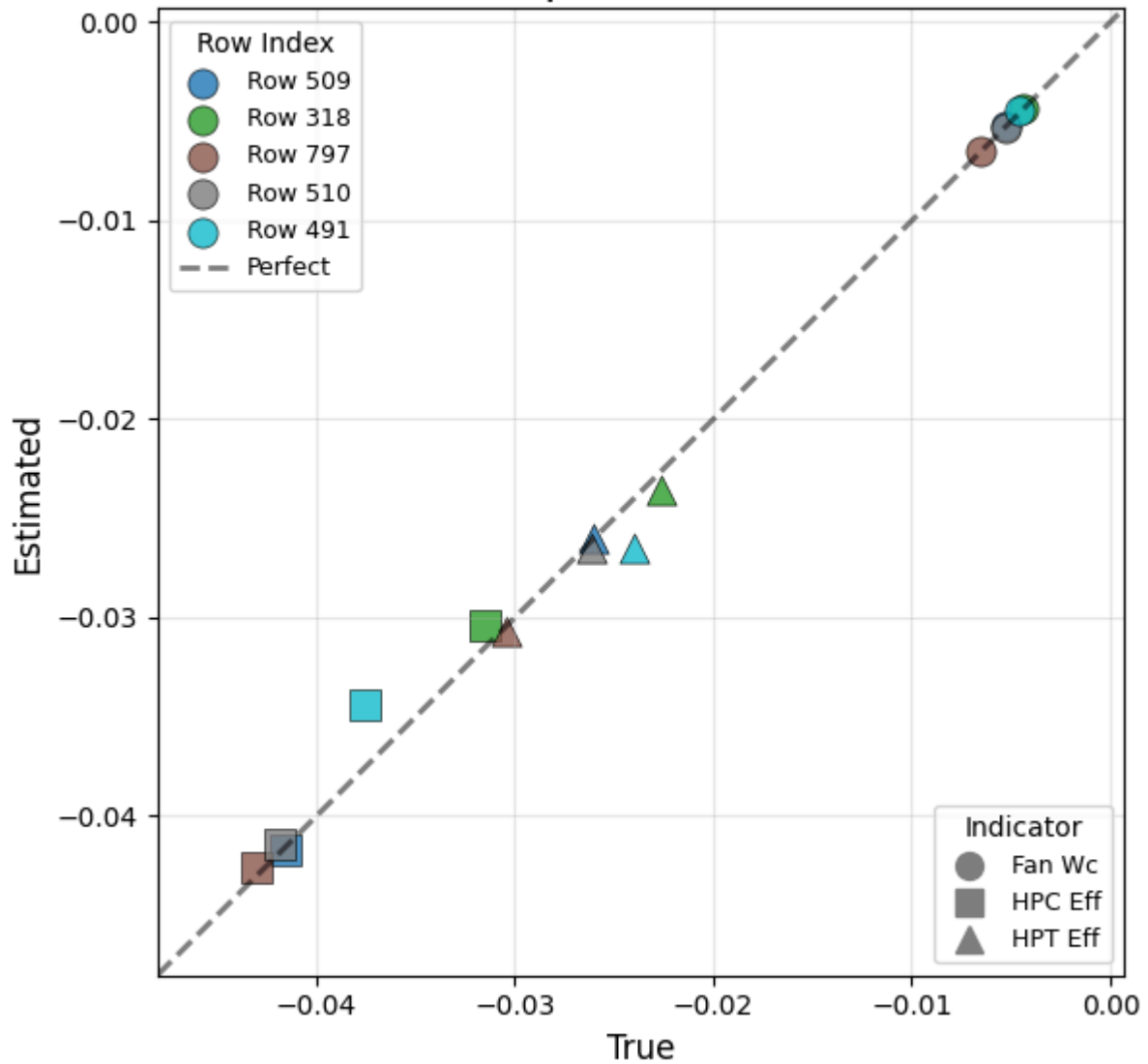
Mean Time: 2578.25s ± 540.88s

1ctx_3sensors: ['TAKEOFF'] × ['HPC_Tin', 'LPT_Tin', 'HPC_Pout_st']
 3 measurements → 3 indicators

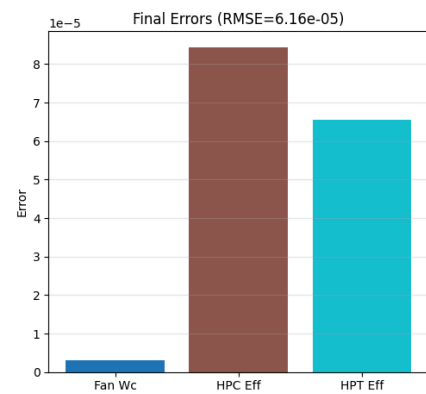
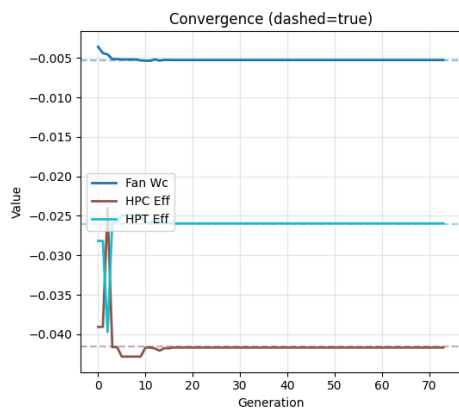
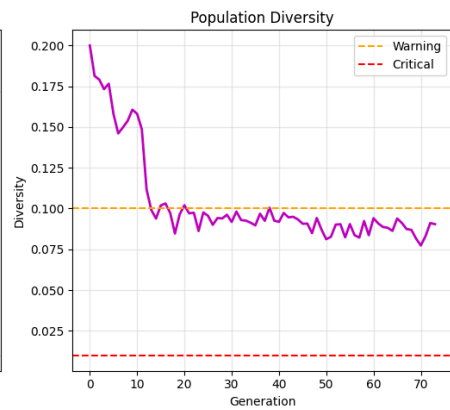
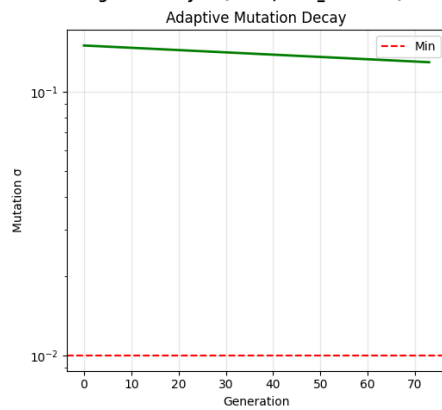
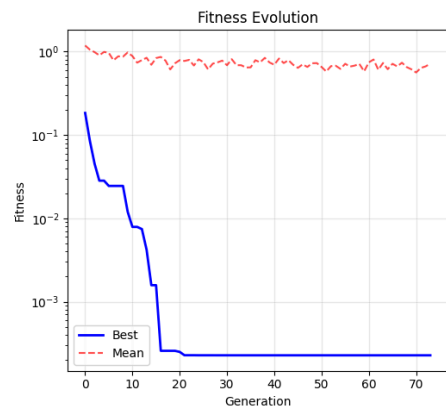
Clean | 1 ctx x 3 sens



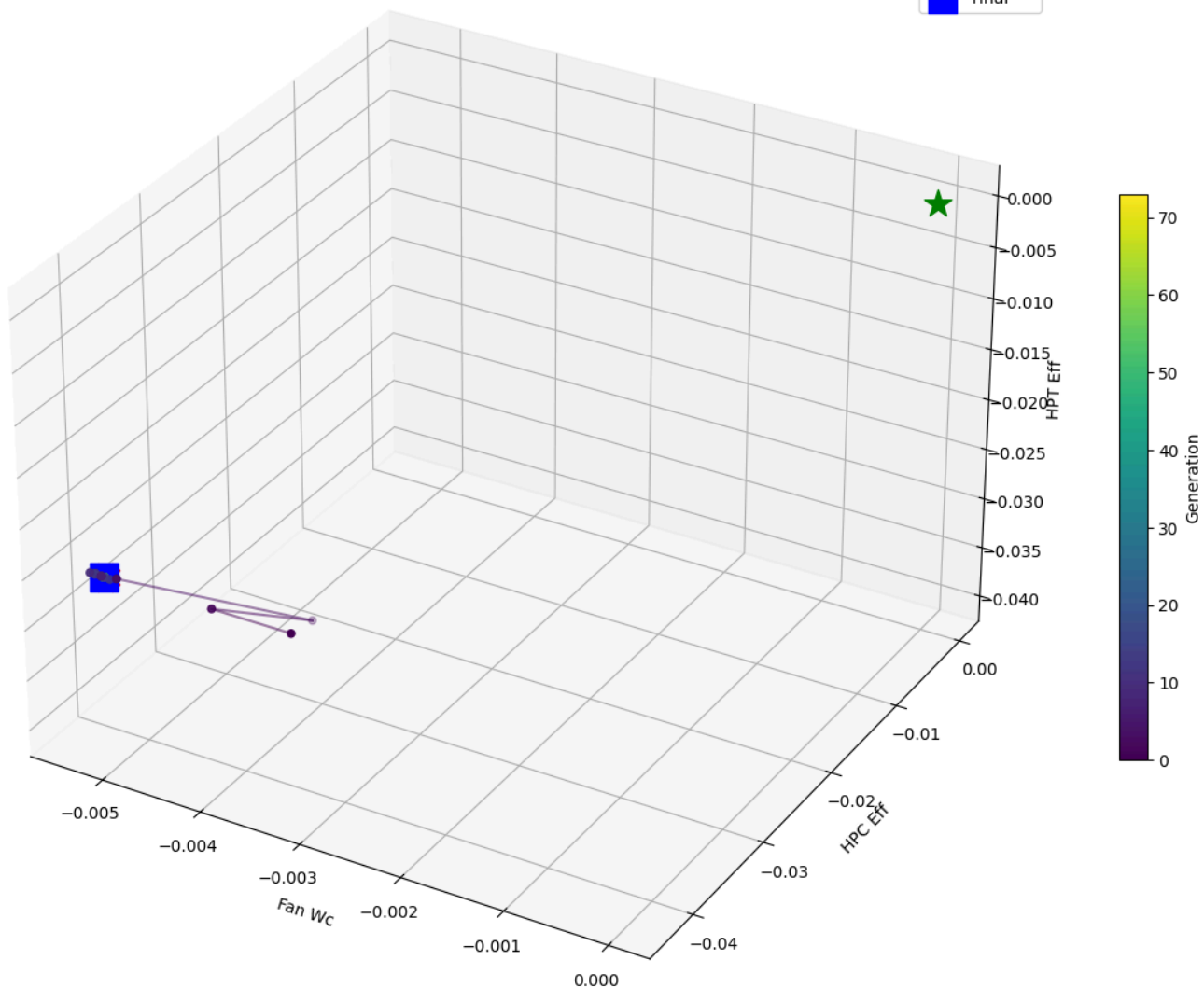
Clean | 1 ctx x 3 sens



Convergence Analysis (clean, 1ctx_3sensors)



3D Search Trajectory (clean, 1ctx_3sensors)



Running ablation study...
Running BASELINE...
RMSE: 1.97e-03

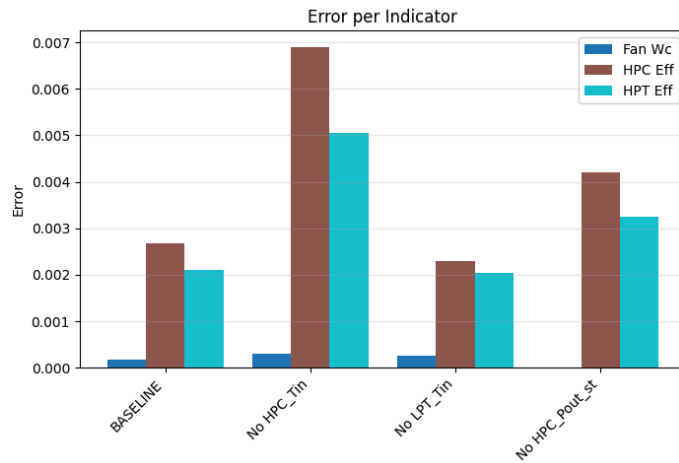
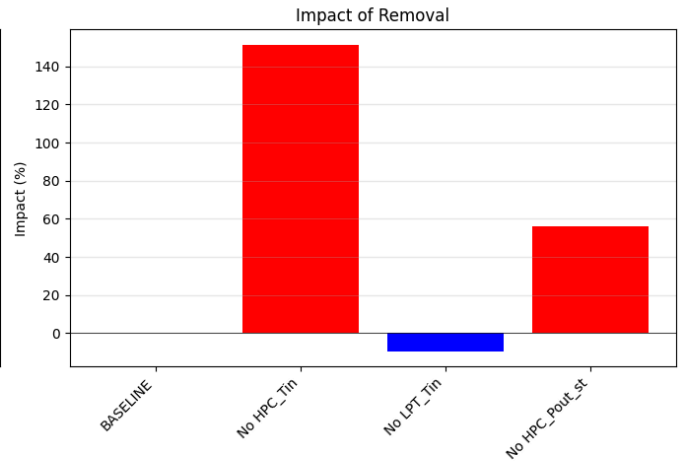
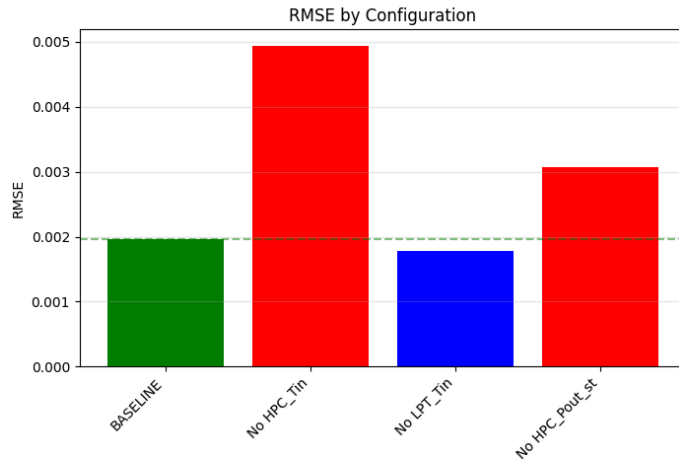
Sensor ablation:

Without HPC_Tin... RMSE: 4.94e-03 (+151.3%)

Without LPT_Tin... RMSE: 1.78e-03 (-9.5%)

Without HPC_Pout_st... RMSE: 3.07e-03 (+56.1%)

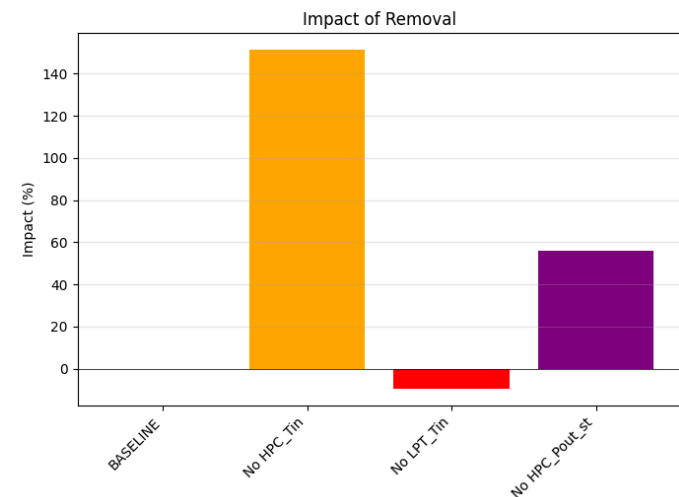
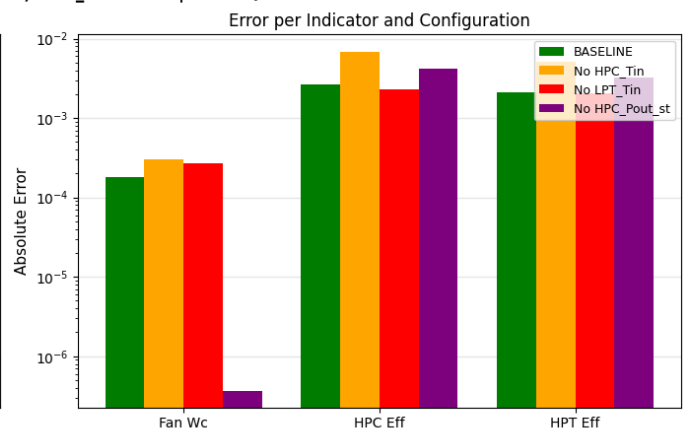
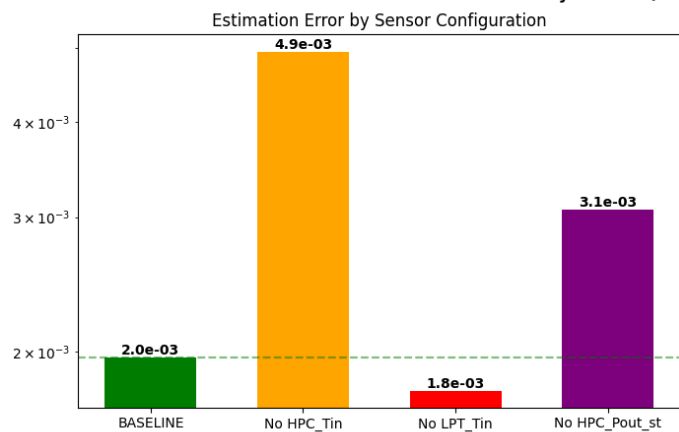
Ablation Study Results (clean, 1ctx_3sensors | row 53)



INTERPRETATION:
BASELINE: RMSE = 1.97e-03
HPC_Tin: HIGH IMPORTANCE (+151.3%)
LPT_Tin: WEAKLY DETRIMENTAL (-9.5%)
HPC_Pout_st: HIGH IMPORTANCE (+56.1%)

Legend:
DETRIMENTAL: Removing HELPS
NEUTRAL: Little effect
HIGH IMPORTANCE: Removing HURTS

Ablation Study Results (clean, 1ctx_3sensors | row 53)



INTERPRETATION:
BASELINE: RMSE = 1.97e-03
HPC_Tin: HIGH IMPORTANCE (+151.3%)
LPT_Tin: WEAKLY DETRIMENTAL (-9.5%)
HPC_Pout_st: HIGH IMPORTANCE (+56.1%)

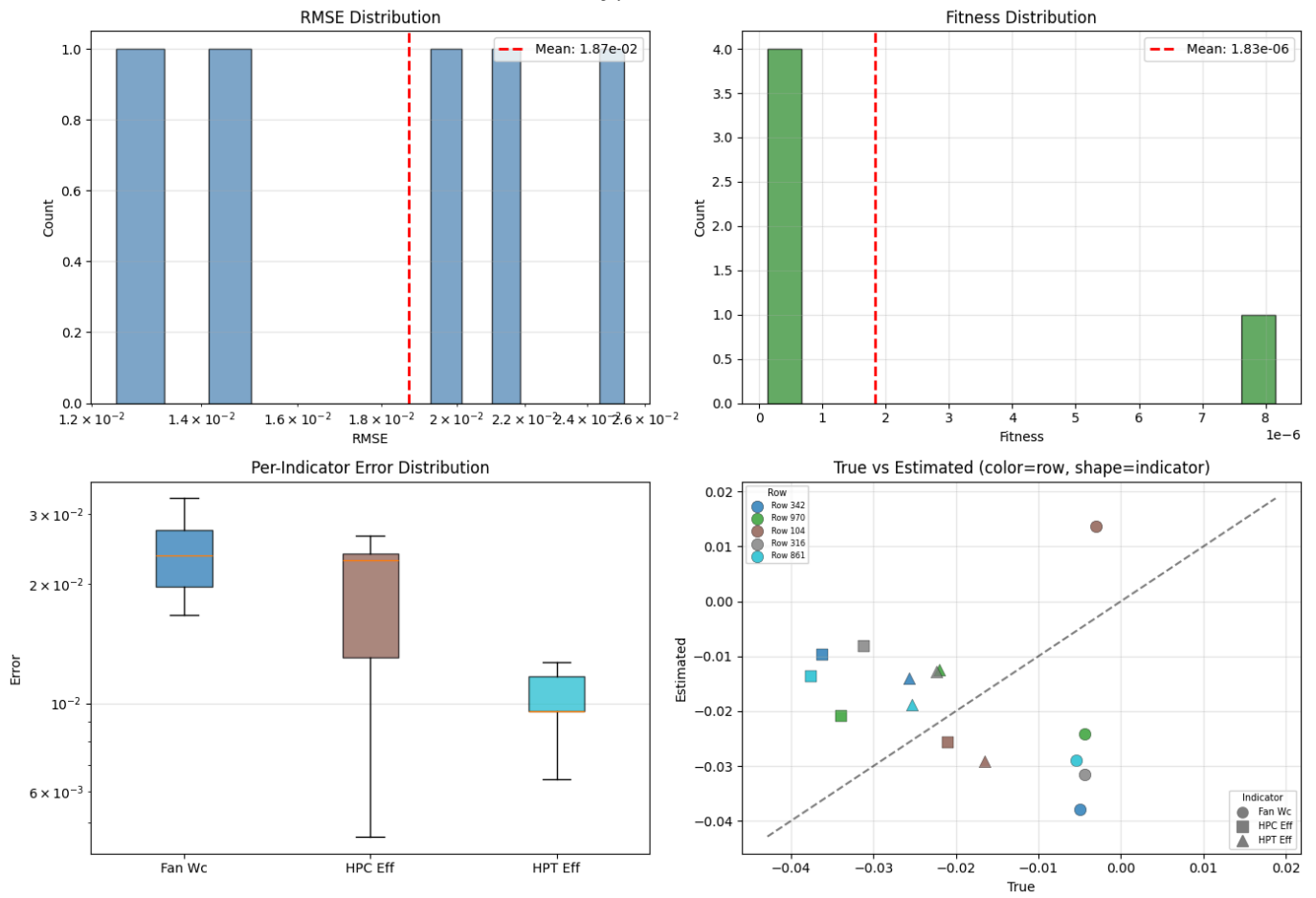
Legend:
DETRIMENTAL: Removing HELPS
NEUTRAL: Little effect
HIGH IMPORTANCE: Removing HURTS

Mean RMSE: $7.66\text{e-}04 \pm 9.10\text{e-}04$
Mean Time: $4223.87\text{s} \pm 2101.94\text{s}$

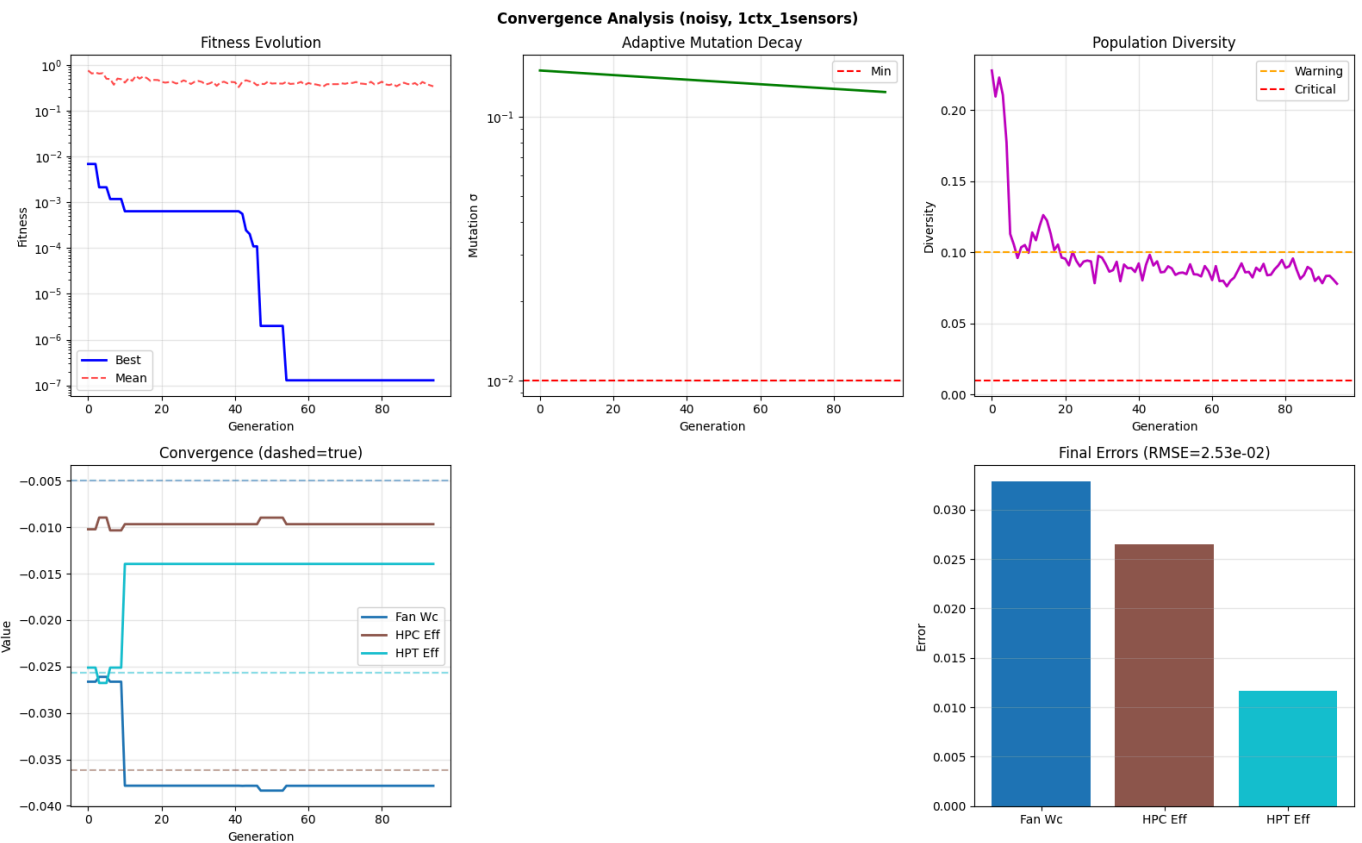
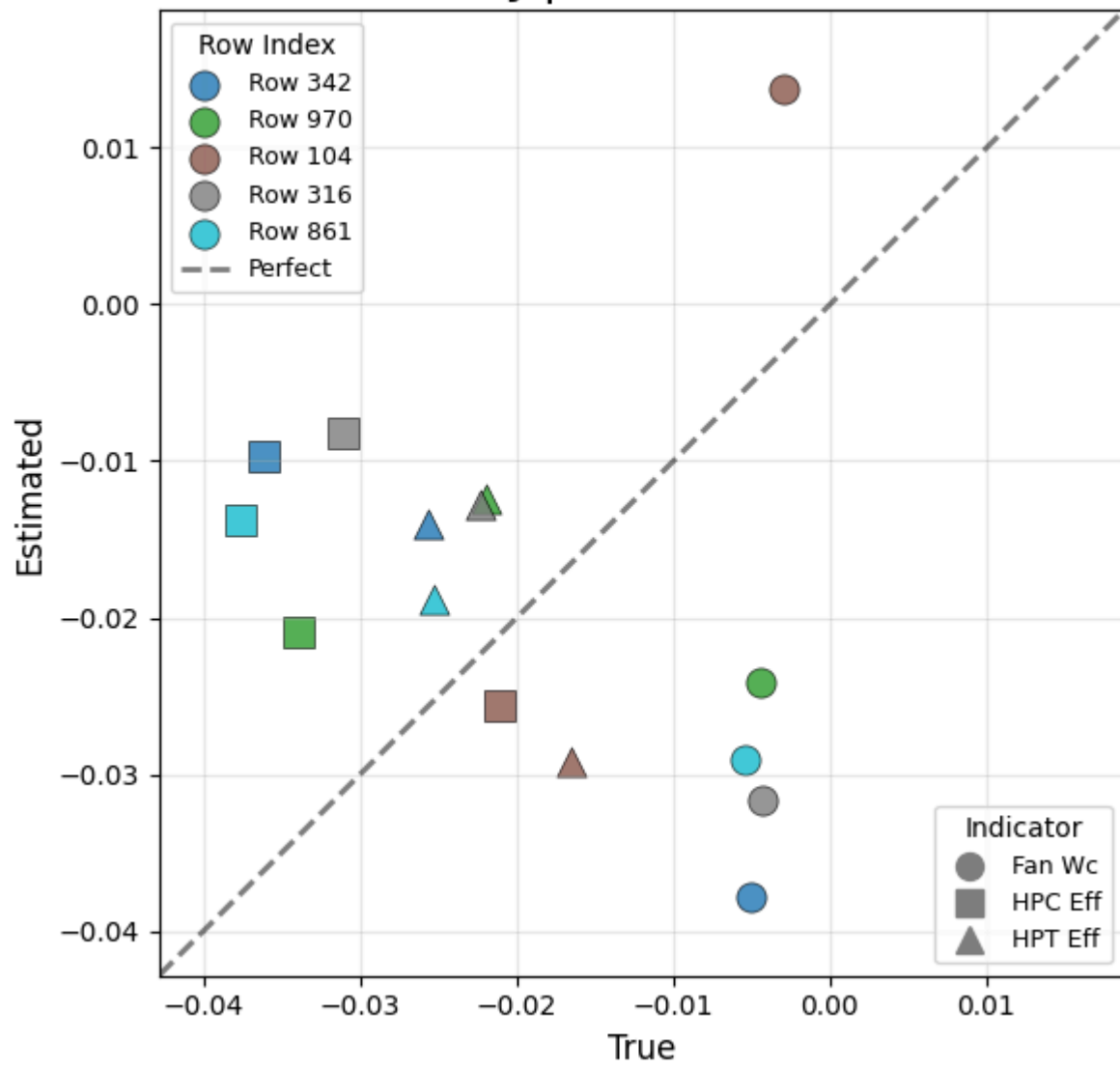
Running experiments on NOISY data

lctx_1sensors: ['TAKEOFF'] × ['HPC_Tin']
1 measurements → 3 indicators

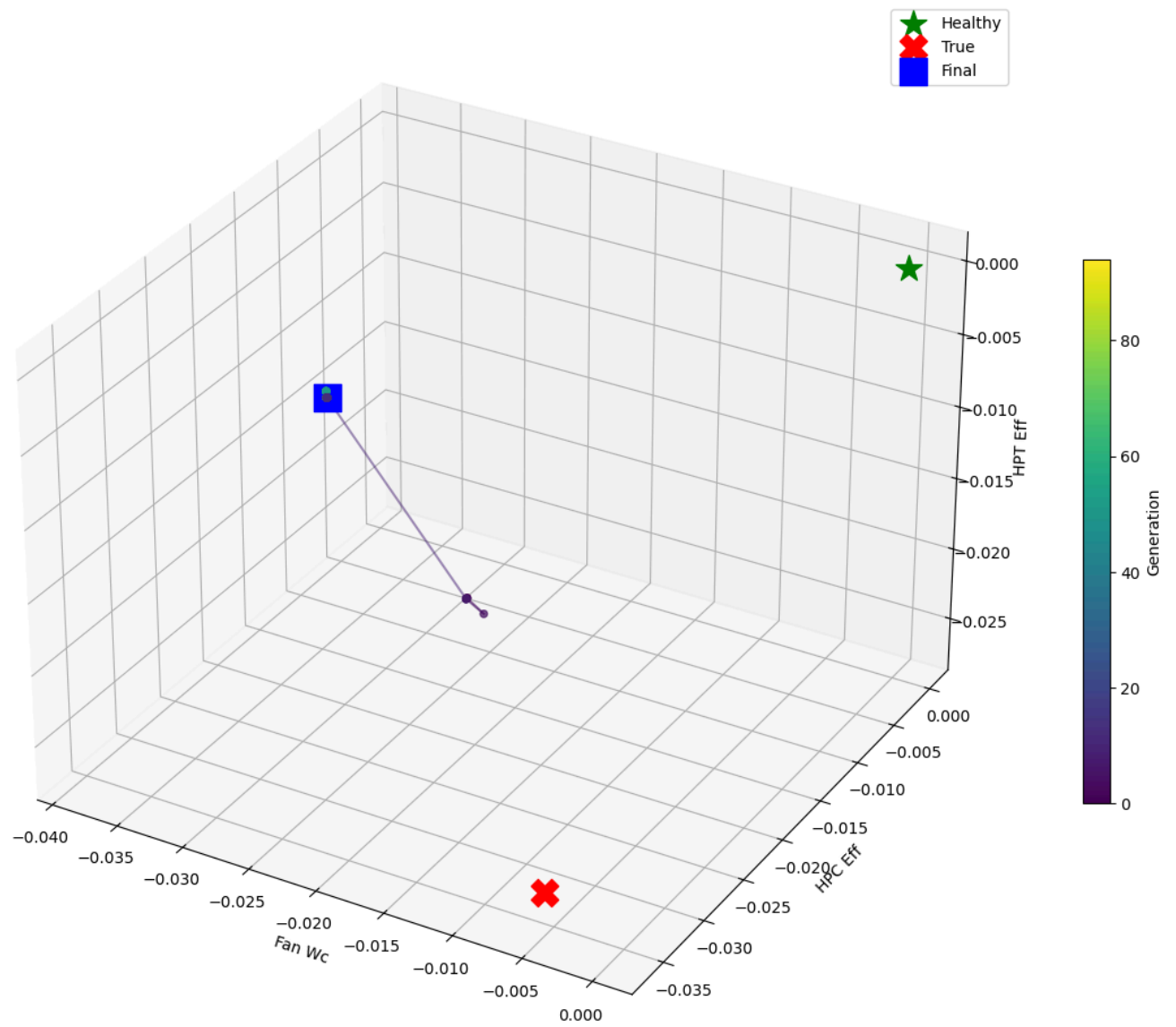
Noisy | 1 ctx × 1 sens



Noisy | 1 ctx × 1 sens



3D Search Trajectory (noisy, 1ctx_1sensors)



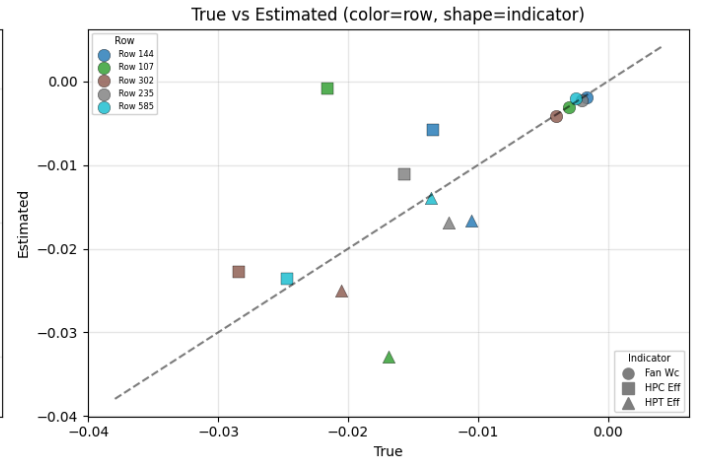
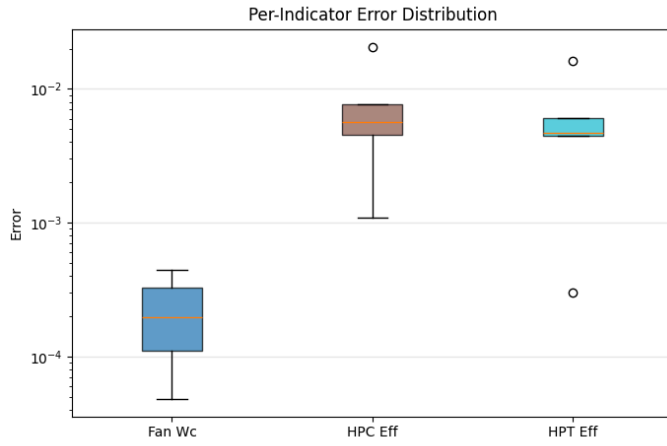
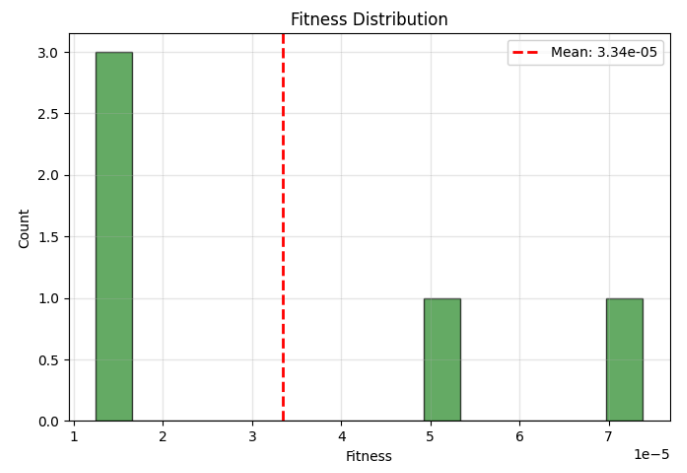
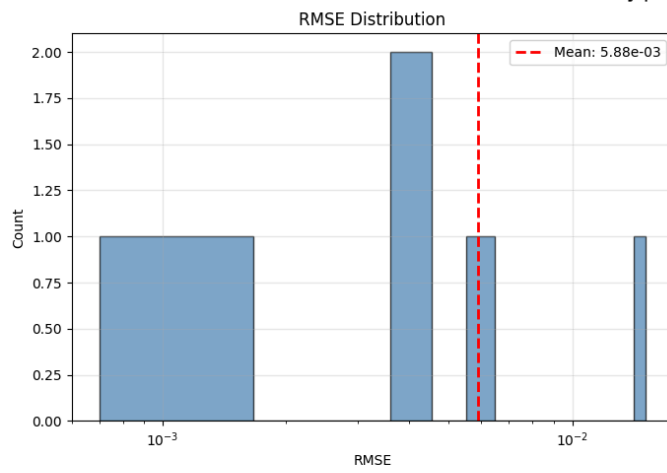
Mean RMSE: $1.87e-02 \pm 5.16e-03$

Mean Time: $1457.31s \pm 361.65s$

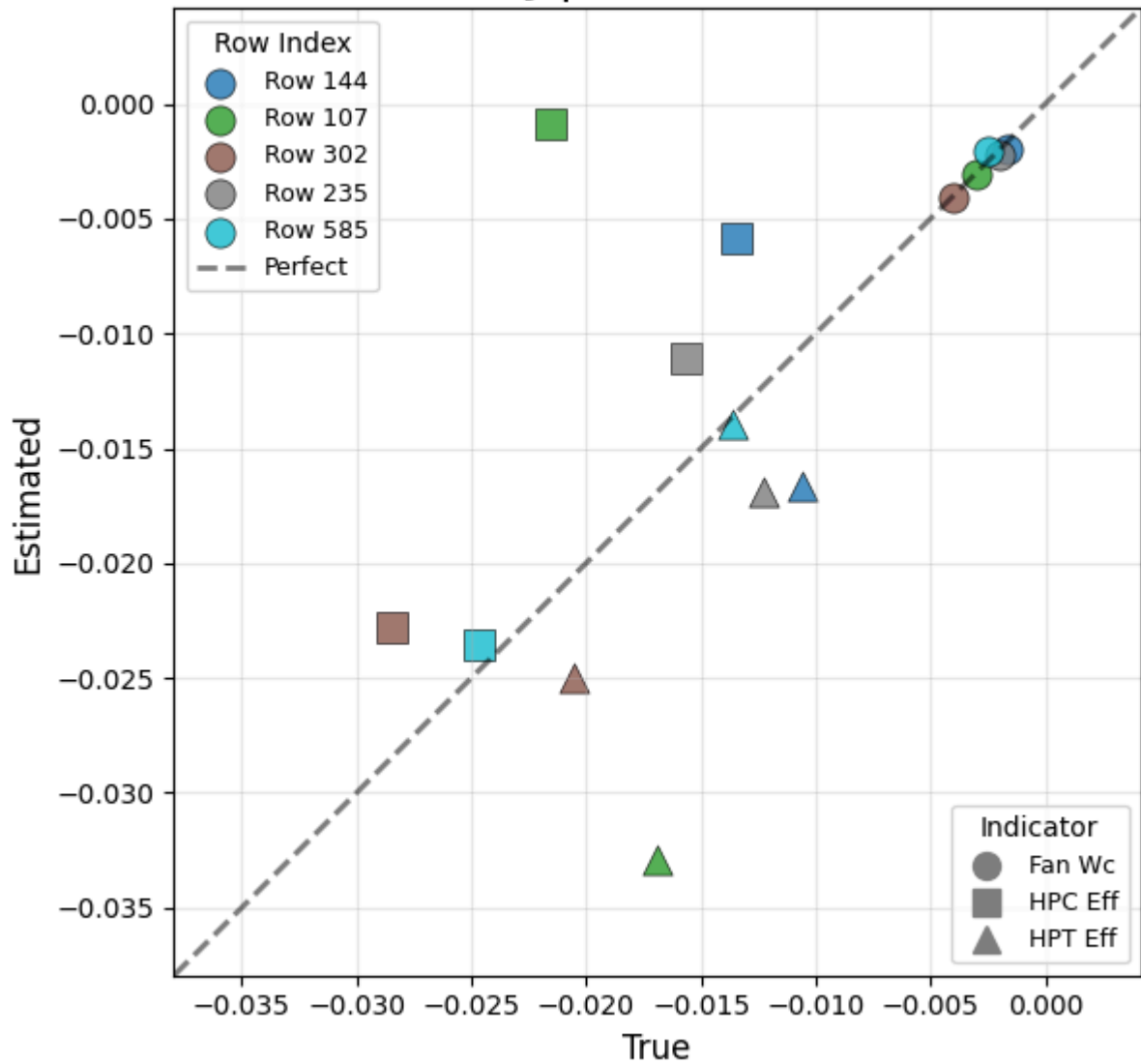
1ctx_2sensors: ['TAKEOFF'] × ['HPC_Tin', 'LPT_Tin']

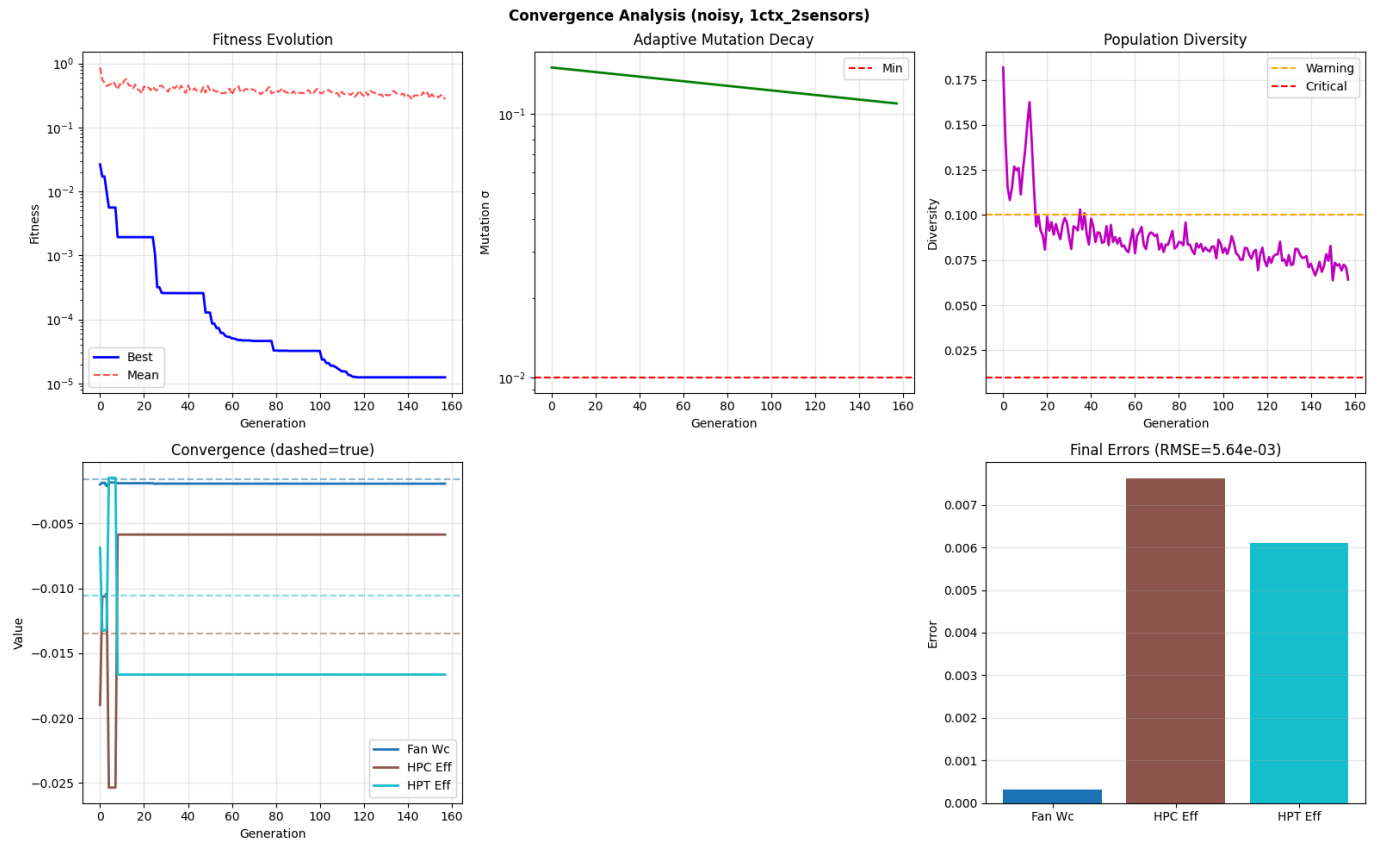
2 measurements → 3 indicators

Noisy | 1 ctx x 2 sens

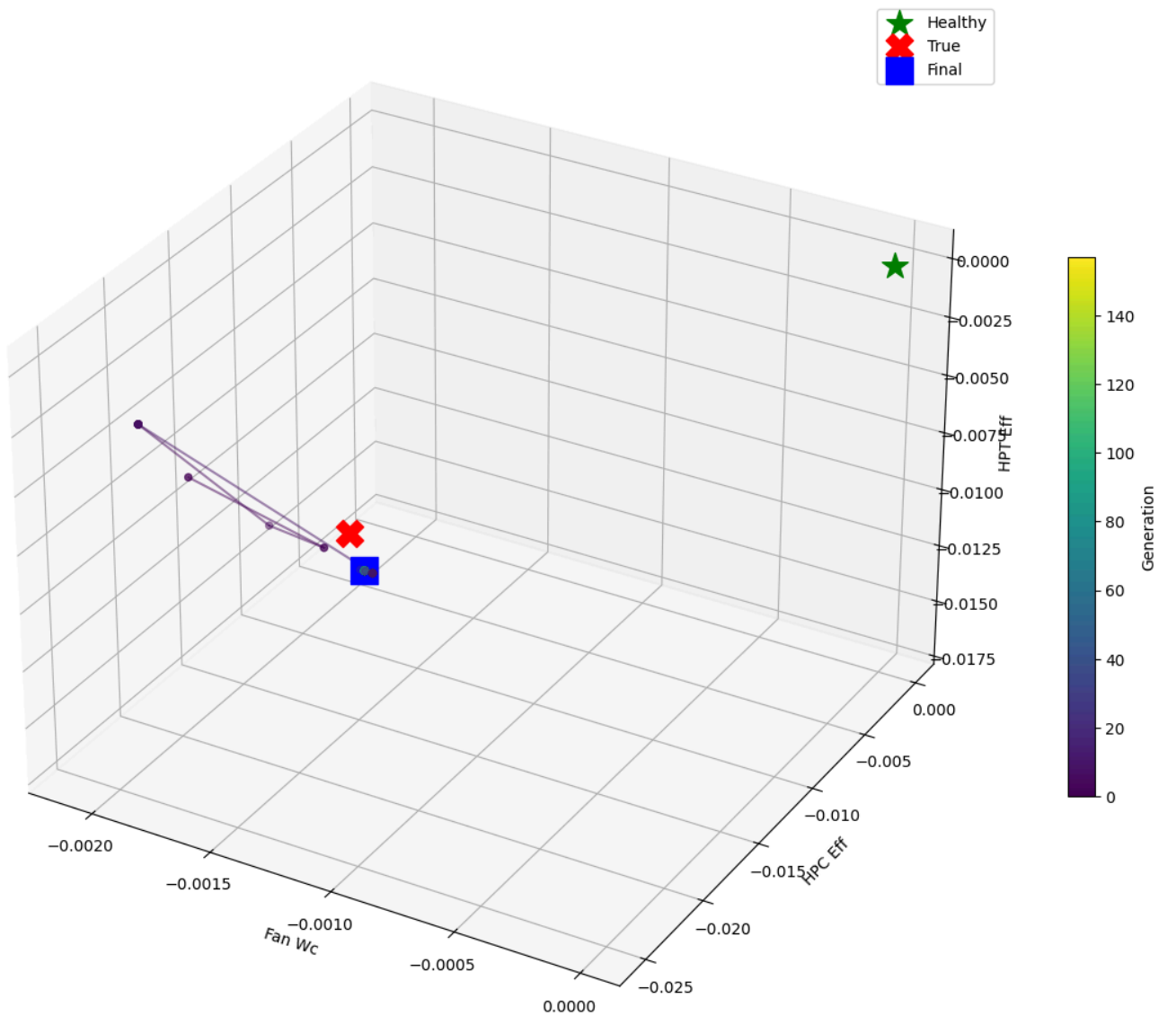


Noisy | 1 ctx x 2 sens



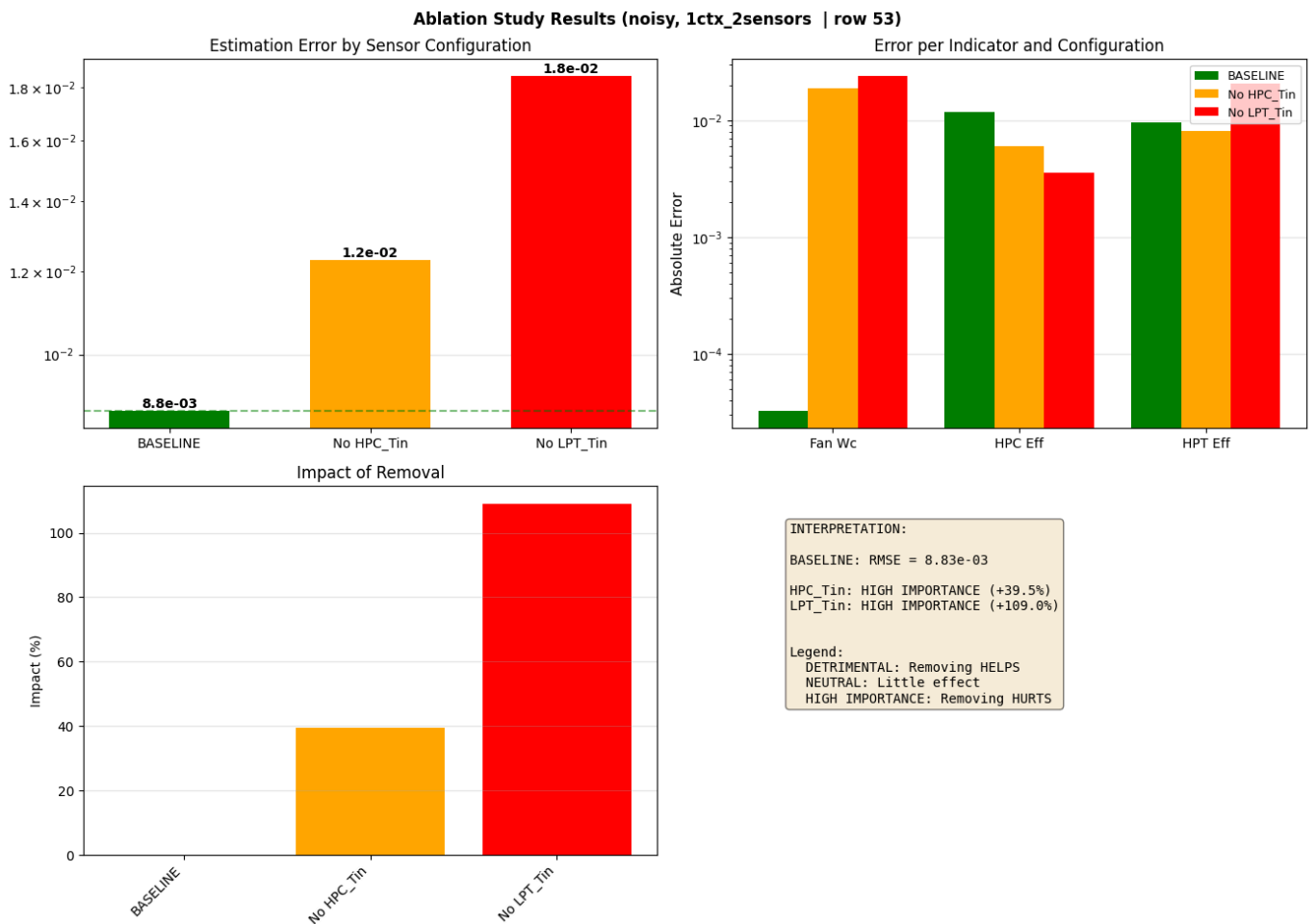
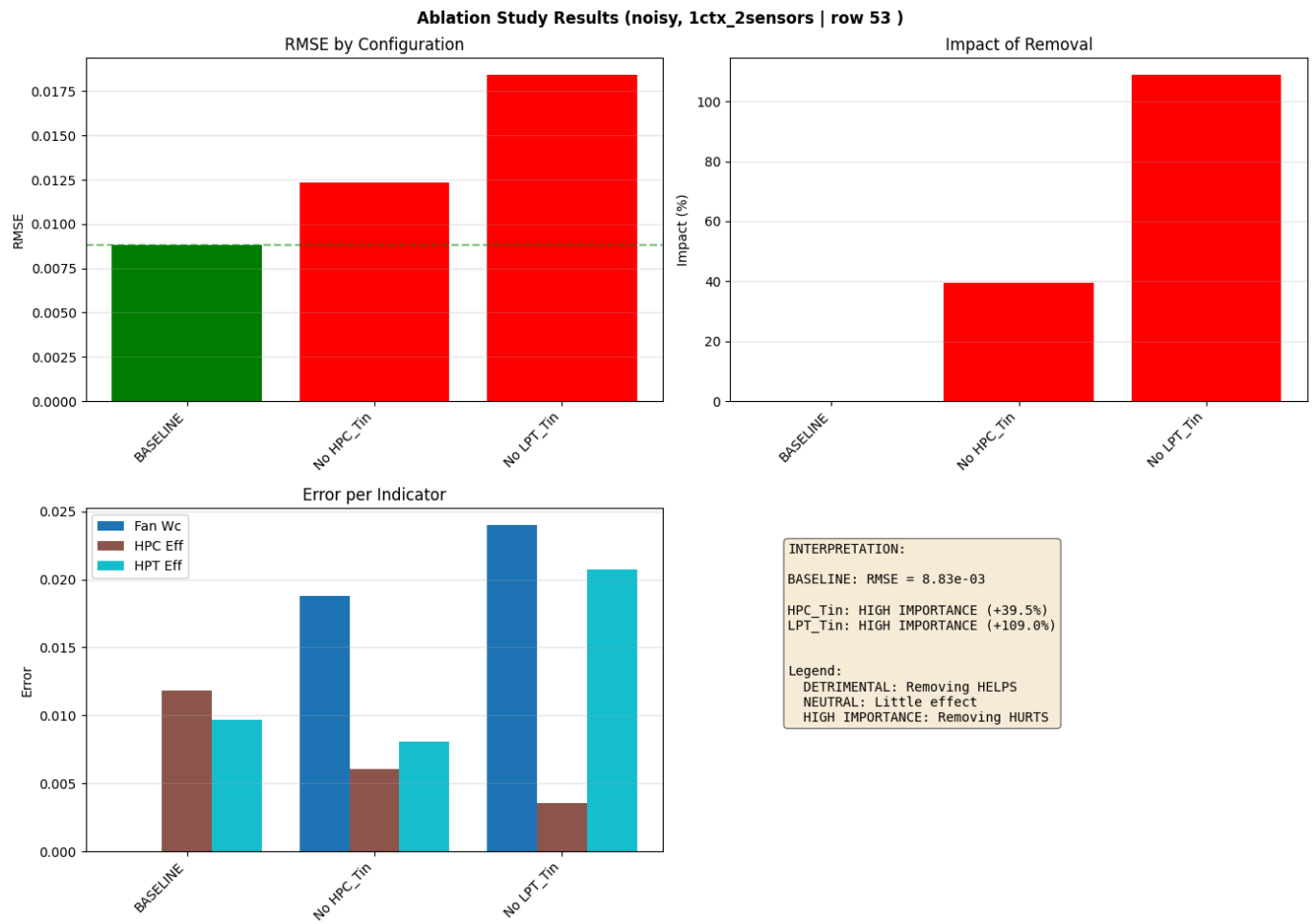


3D Search Trajectory (noisy, 1ctx_2sensors)



Running ablation study...
Running BASELINE...
RMSE: 8.83e-03

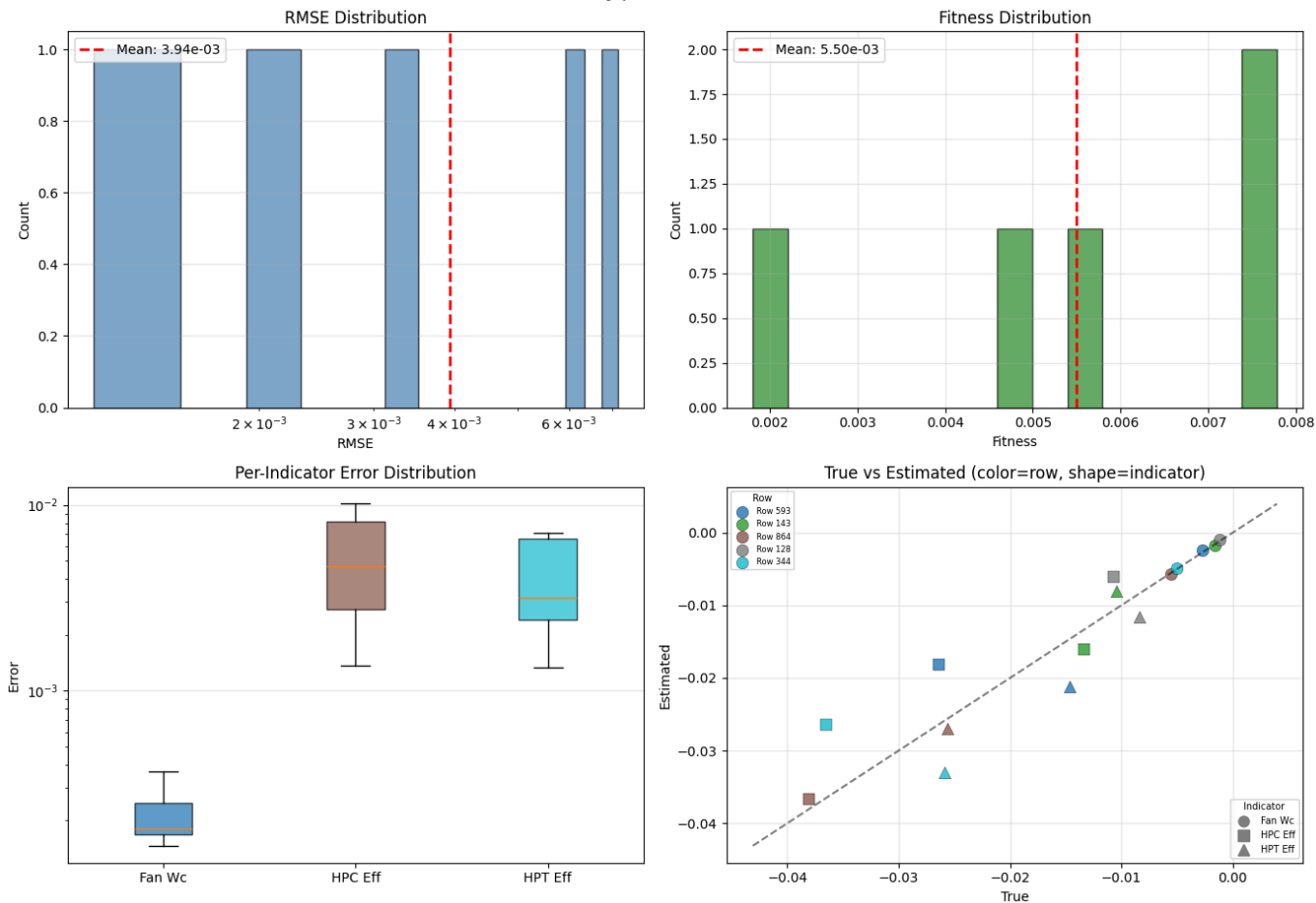
Sensor ablation:
Without HPC_Tin... RMSE: 1.23e-02 (+39.5%)
Without LPT_Tin... RMSE: 1.85e-02 (+109.0%)



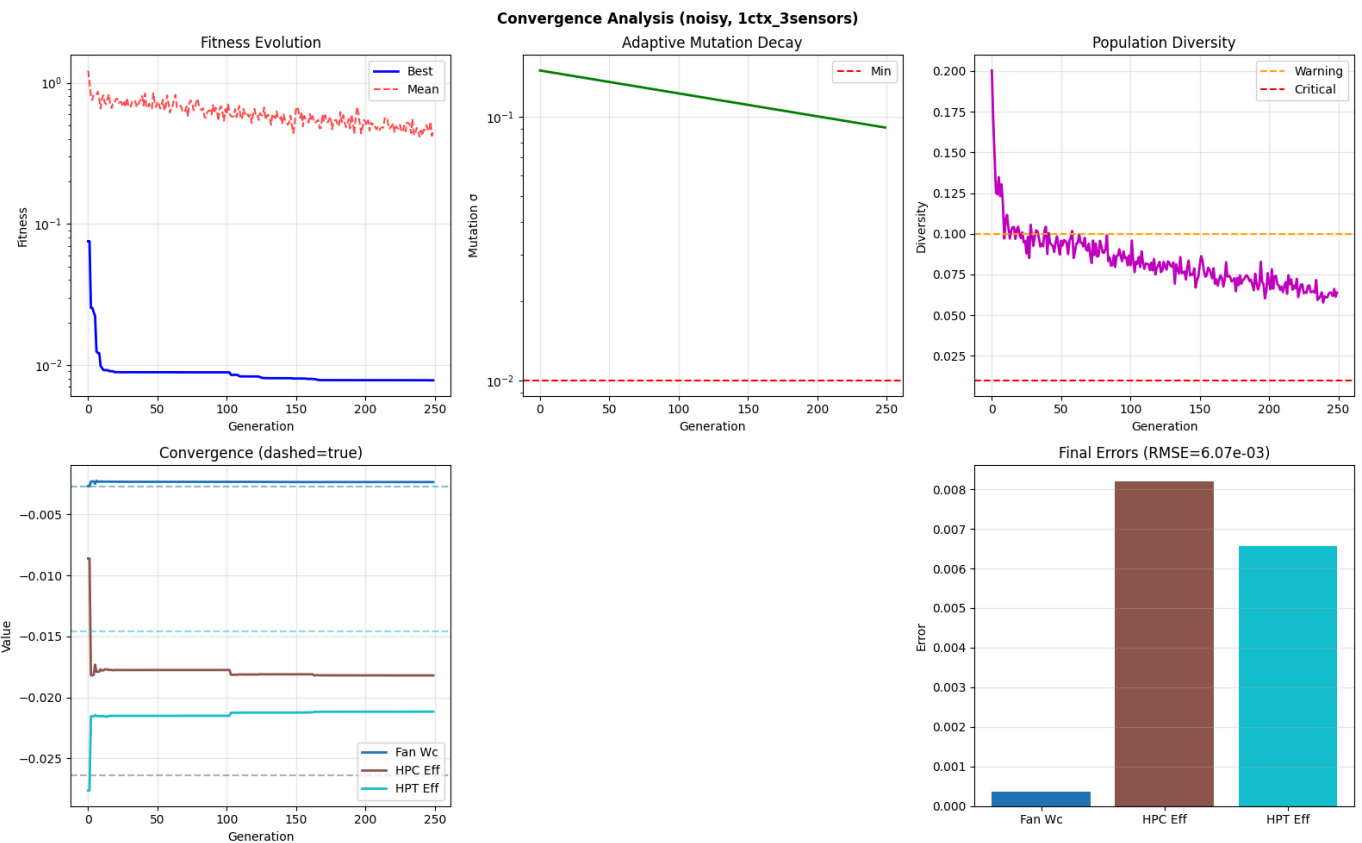
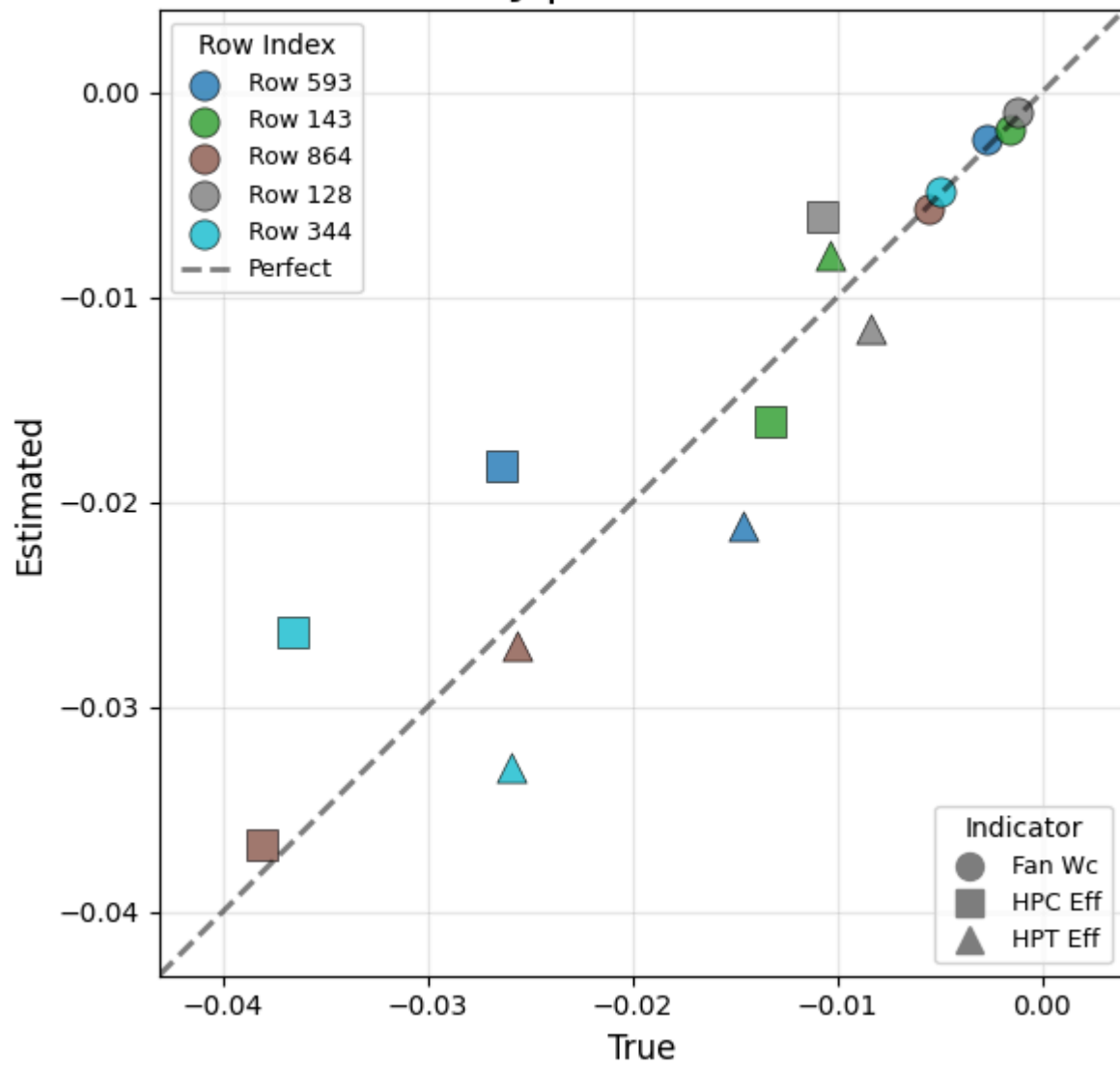
Mean RMSE: $5.88\text{e-}03 \pm 5.48\text{e-}03$
Mean Time: 2534.88s $\pm$ 951.90s

lctx_3sensors: ['TAKEOFF'] $\times$ ['HPC_Tin', 'LPT_Tin', 'HPC_Pout_st']
3 measurements $\rightarrow$ 3 indicators

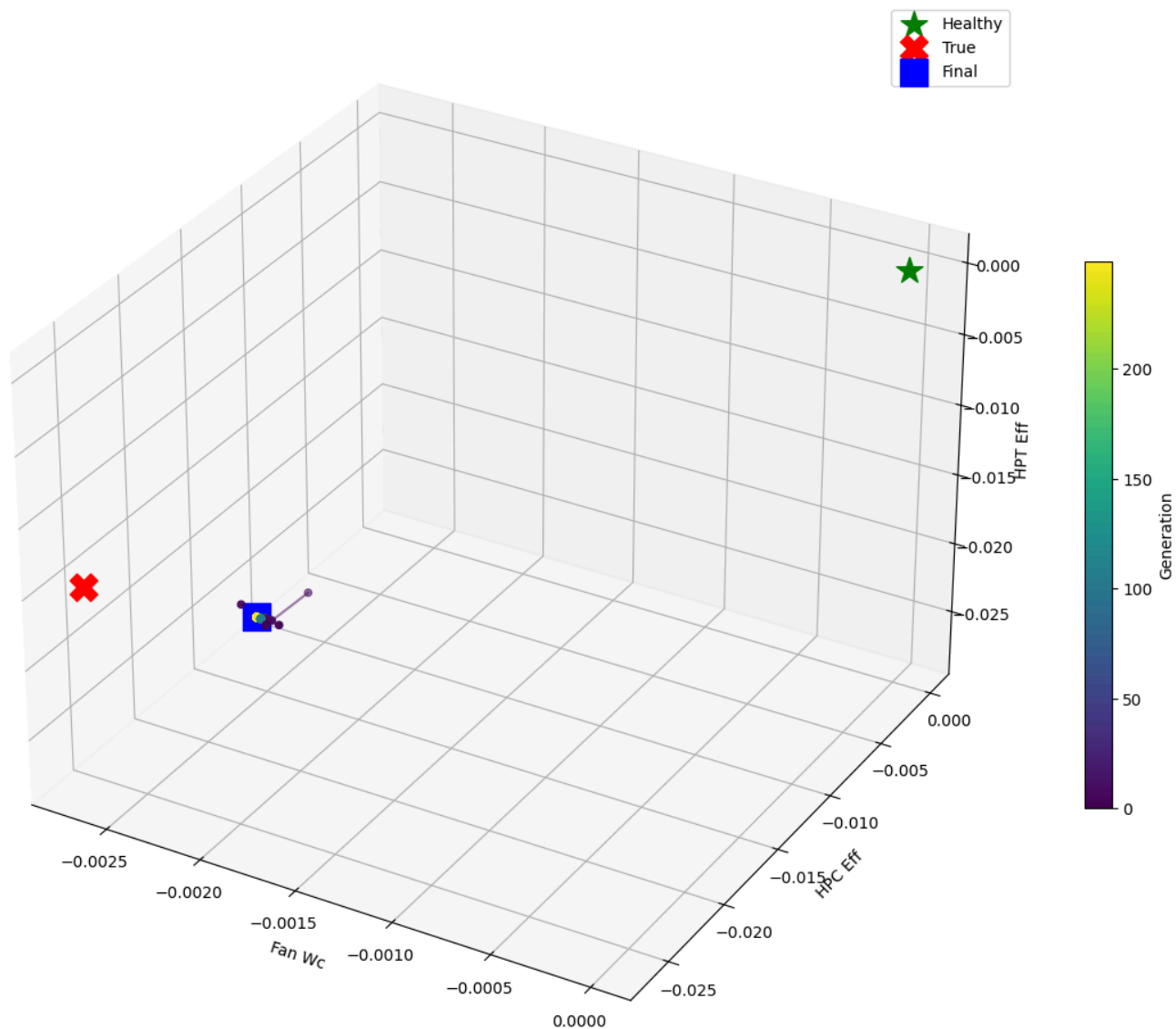
Noisy | 1 ctx $\times$ 3 sens



Noisy | 1 ctx × 3 sens



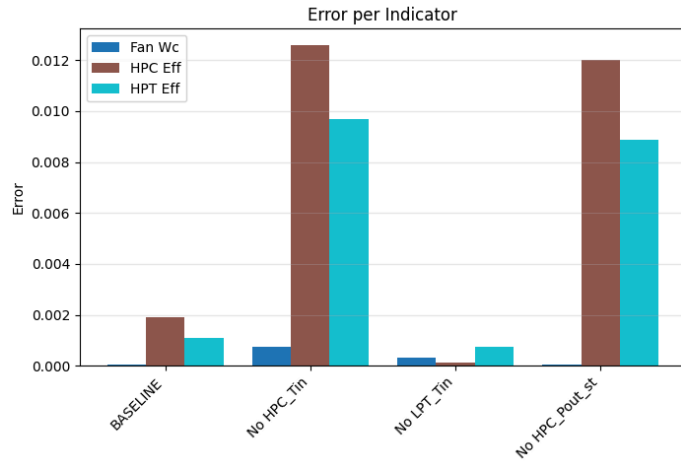
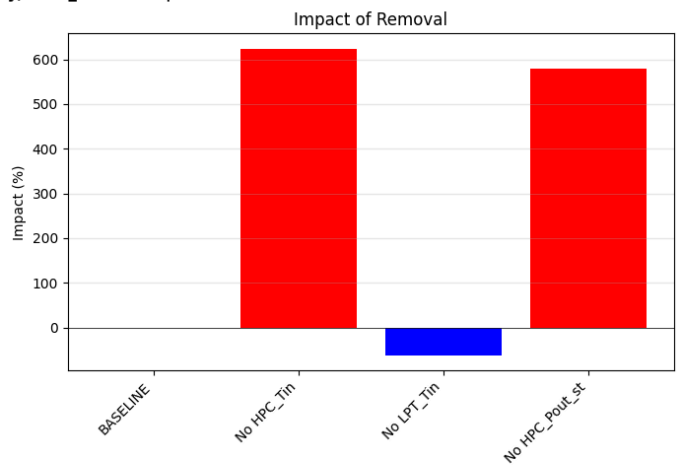
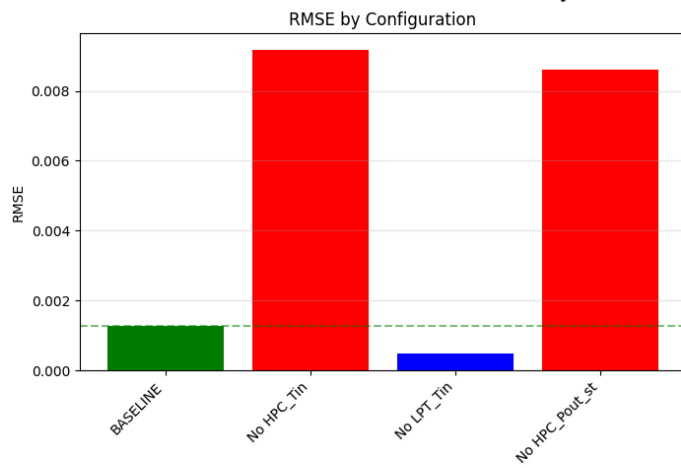
3D Search Trajectory (noisy, 1ctx_3sensors)



Running ablation study...
Running BASELINE...
RMSE: 1.27e-03

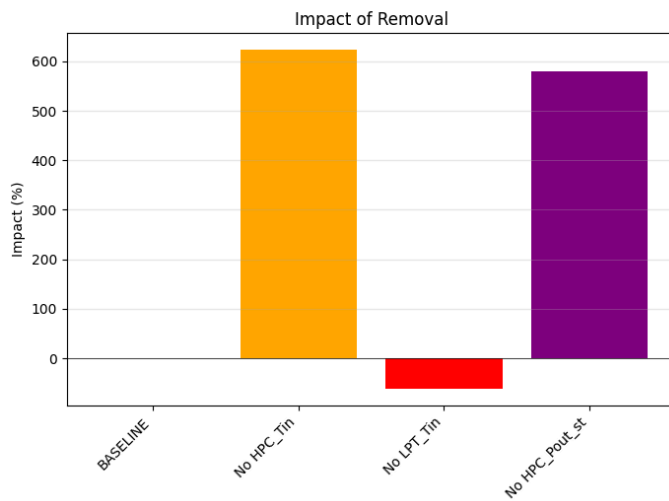
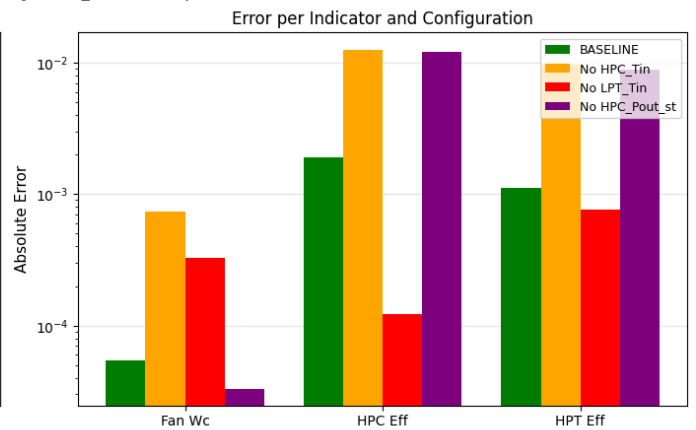
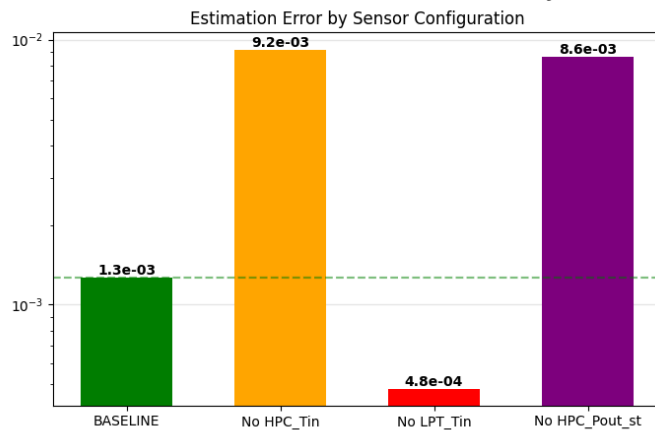
Sensor ablation:
Without HPC_Tin... RMSE: 9.18e-03 (+623.6%)
Without LPT_Tin... RMSE: 4.83e-04 (-62.0%)
Without HPC_Pout_st... RMSE: 8.62e-03 (+579.4%)

Ablation Study Results (noisy, 1ctx_3sensors | row 53)



INTERPRETATION:
 BASELINE: RMSE = 1.27e-03
 HPC_Tin: HIGH IMPORTANCE (+623.6%)
 LPT_Tin: DETRIMENTAL (-62.0%)
 HPC_Pout_st: HIGH IMPORTANCE (+579.4%)
 Legend:
 DETRIMENTAL: Removing HELPS
 NEUTRAL: Little effect
 HIGH IMPORTANCE: Removing HURTS

Ablation Study Results (noisy, 1ctx_3sensors | row 53)



INTERPRETATION:
 BASELINE: RMSE = 1.27e-03
 HPC_Tin: HIGH IMPORTANCE (+623.6%)
 LPT_Tin: DETRIMENTAL (-62.0%)
 HPC_Pout_st: HIGH IMPORTANCE (+579.4%)
 Legend:
 DETRIMENTAL: Removing HELPS
 NEUTRAL: Little effect
 HIGH IMPORTANCE: Removing HURTS

Mean RMSE: 3.94e-03 ± 2.57e-03

Mean Time: 4649.16s ± 1352.48s

CPU times: user 1h 5min 28s, sys: 1min 44s, total: 1h 7min 12s

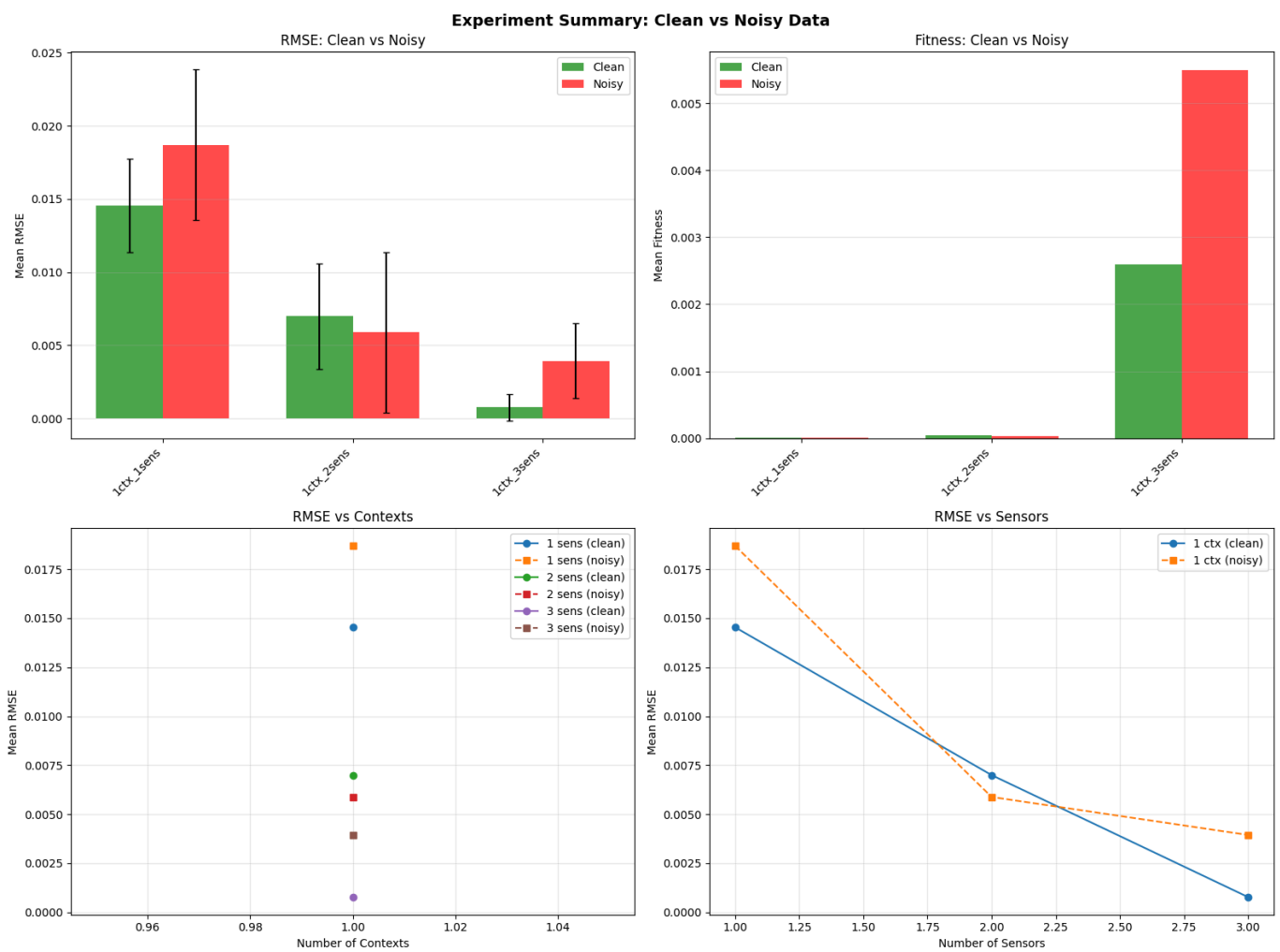
Wall time: 1d 9h 18min 10s

In []:

```
In [ ]:
In [ ]:
In [ ]:
In [ ]:
In [ ]:
In [ ]:
In [ ]:
In [ ]:
```

4. Summary Comparison Plot

```
In [18]: %%time
# Generate summary plot
plot_summary_comparison(summary_df, save_path=os.path.join(output_dir, "summary_compa
```



CPU times: user 800 ms, sys: 17 ms, total: 817 ms
Wall time: 815 ms

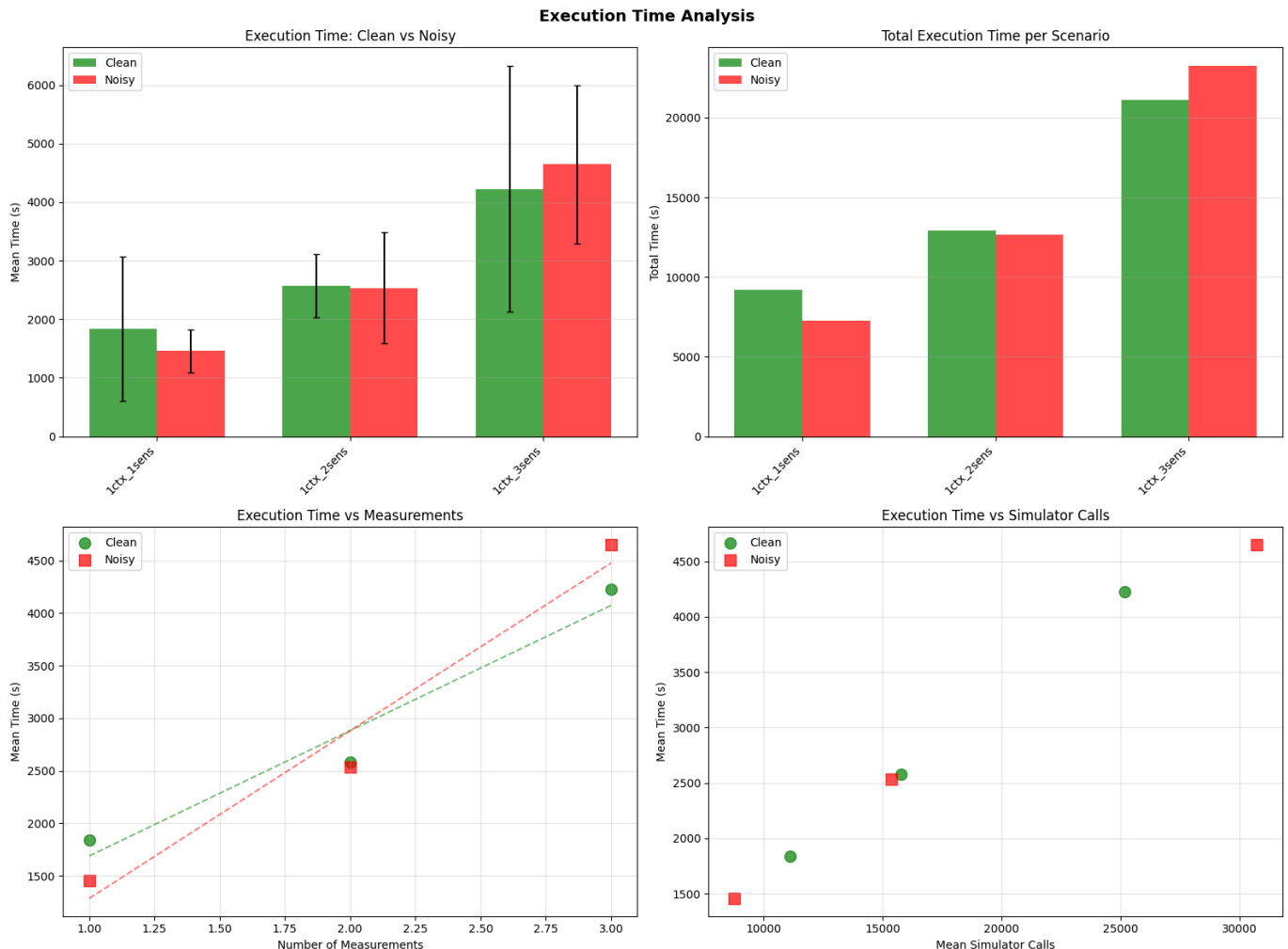
```
In [ ]:
In [ ]:
```

In []:

In []:

5. PLOT exucution time

```
In [19]: %%time
# Generate execution time plot
plot_execution_times(summary_df, save_path=os.path.join(output_dir, "execution_time_a
print(f"Execution time plot saved to: {os.path.join(output_dir, 'execution_time_analy
```



Execution time plot saved to: ../experiments/run_20260127_060325/execution_time_analysis.png

CPU times: user 771 ms, sys: 18 ms, total: 789 ms

Wall time: 788 ms

6. Summary of the experiemnt

Summary of the results over multiple index row

```
In [20]: # Display summary table
print("\n" + "="*100)
print("EXPERIMENT SUMMARY")
print("="*100)
print(summary_df[['data_type', 'n_contexts', 'n_sensors', 'n_measurements',
                  'mean_rmse', 'std_rmse', 'mean_fitness', 'mean_time', 'std_time',
                  'mean_sim_calls', 'mean_generations']].to_string(index=False))

# Print timing summary
print("\n" + "="*100)
print("TIMING SUMMARY")
```



```

print("="*100)
clean_total = summary_df[summary_df['data_type']=='clean']['total_time'].sum()
noisy_total = summary_df[summary_df['data_type']=='noisy']['total_time'].sum()
print(f"Total time for CLEAN experiments: {clean_total:.1f}s ({clean_total/60:.1f} mi
print(f"Total time for NOISY experiments: {noisy_total:.1f}s ({noisy_total/60:.1f} mi
print(f"Grand total: {clean_total + noisy_total:.1f}s ({(clean_total + noisy_total)/6

```

===== ===== EXPERIMENT SUMMARY ===== =====

data_type	n_contexts	n_sensors	n_measurements	mean_rmse	std_rmse	mean_fitness	
mean_time	std_time	mean_sim_calls	mean_generations				
clean	1	1	1	0.014538	0.003208	0.000006	18
39.856969	1237.304207	11115.8	83.6				
clean	1	2	2	0.006993	0.003595	0.000040	25
78.247420	540.880370	15794.0	117.4				
clean	1	3	3	0.000766	0.000910	0.002602	42
23.872005	2101.941190	25178.2	184.0				
noisy	1	1	1	0.018693	0.005156	0.000002	14
57.306863	361.654122	8795.6	65.0				
noisy	1	2	2	0.005877	0.005479	0.000033	25
34.882778	951.903678	15384.8	113.0				
noisy	1	3	3	0.003938	0.002572	0.005495	46
49.164880	1352.479808	30710.8	223.0				

===== ===== TIMING SUMMARY ===== =====

Total time for CLEAN experiments: 43209.9s (720.2 min)

Total time for NOISY experiments: 43206.8s (720.1 min)

Grand total: 86416.7s (1440.3 min)

```

In [21]: # Print output directory structure
print(f"\n\nAll results saved in: {output_dir}")
print("\nDirectory structure:")
for root, dirs, files in os.walk(output_dir):
    level = root.replace(output_dir, '').count(os.sep)
    indent = ' ' * 2 * level
    print(f"{indent}{os.path.basename(root)}/")
    subindent = ' ' * 2 * (level + 1)
    for file in files[:5]: # Limit files shown
        print(f"{subindent}{file}")
    if len(files) > 5:
        print(f"{subindent}... and {len(files)-5} more files")

```

All results saved in: ../experiments/run_20260127_060325

Directory structure:

```
run_20260127_060325/  
  execution_time_analysis.png  
  summary_comparison.png  
  experiment_summary.csv  
  clean/  
    lctx_1sensors/  
      trajectory_3d.png  
      multiple_solutions.csv  
      convergence.png  
      results_summary.png  
      trajectory_3d.html  
      ... and 2 more files  
    lctx_3sensors/  
      trajectory_3d.png  
      ablation.csv  
      multiple_solutions.csv  
      convergence.png  
      results_summary.png  
      ... and 5 more files  
    lctx_2sensors/  
      trajectory_3d.png  
      ablation.csv  
      multiple_solutions.csv  
      convergence.png  
      results_summary.png  
      ... and 5 more files  
  noisy/  
    lctx_1sensors/  
      trajectory_3d.png  
      multiple_solutions.csv  
      convergence.png  
      results_summary.png  
      trajectory_3d.html  
      ... and 2 more files  
    lctx_3sensors/  
      trajectory_3d.png  
      ablation.csv  
      multiple_solutions.csv  
      convergence.png  
      results_summary.png  
      ... and 5 more files  
    lctx_2sensors/  
      trajectory_3d.png  
      ablation.csv  
      multiple_solutions.csv  
      convergence.png  
      results_summary.png  
      ... and 5 more files
```

In []: